

# MIRACLE simulation outputs metadata specification version 3.0

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## Abstract

This document contains metadata specifications for recording simulation outputs.

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## 1 ChangeLog

| Version | Date          | Person                                                                      | Description                                                                                                                                                                                                                                  |
|---------|---------------|-----------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1.1.3   | 28 April 2015 | Lorenzo Milazzo and Gary Polhill                                            | Original version                                                                                                                                                                                                                             |
| 1.1.4   | 18 June 2015  | Gary Polhill, Lorenzo Milazzo, Dawn Parker, Calvin Pritchard, Xiongbing Jin | Enhancements for workflow specifications and links with a metadata schema from earlier work at the University of Waterloo                                                                                                                    |
| 1.1.5   | 7 July 2015   | Gary Polhill and Lorenzo Milazzo                                            | Improvements to fine grain, assumptions and statistics; explicit links to possible external ontologies; record of visualisations                                                                                                             |
| 1.1.6   | 26 Nov 2015   | Ju-Sung Lee and Gary Polhill                                                | Improvements to the fine grain; cosmetic improvements                                                                                                                                                                                        |
| 1.1.7   | 29 Apr 2016   | Doug Salt                                                                   | Corrections. Add Description field to everything. Add About field to everything. Remove Purpose from Pipeline as redundant. Remove Purpose from Application as redundant. Change statisticalInput, statisticalVariable and statisticalMethod |
| 2.0.0   | 8 Nov 2022    | Doug Salt                                                                   | Tidied up the relations                                                                                                                                                                                                                      |
| 3.0.0   | 9 Sep 2026    | Doug Salt                                                                   | Converted to markdown and updated for RESAS release                                                                                                                                                                                          |

## 2 Introduction

The replication ‘crisis’ (Fanelli 2018) in science generally (and especially the social sciences (Shrout and Rodgers 2018)) has its correlate in social simulation (Edmonds and Hales 2003). In our own work (Polhill et al. 2017), the complicated computational workflows associated with preparing, running and analysing the results of agent-based models have demonstrated the need for automated record-keeping.

Earlier work identified the various ways in which provenance metadata could be used to help analyse agent-based models (Pignotti et al. 2013). Three types of provenance were identified in that work:

1. The social processes outlining the history of model development.
2. Execution of the model and associated code for analysing its results.
3. Detailed state-change metadata per run.

This document provides a framework in which the first two of these types of provenance can be recorded.

## 3 Overview

This specification builds on earlier work towards metadata standards for sharing simulation outputs (Polhill et al. 2014). The schema diagram is shown in fig. 2.

Note originally this schema was designed to run on a relational database, but later work has adapted it to run on graph database as well, due to ease of querying on a graph database (Davoudian and Liu 2020). As such there are many reified relationships in the relational description of the database. Please note that in the graph database, the reified relationships have been converted to direct links between entities; the many to many relation being more natural in a graph database setting. A note has been made when a reified relationship in the relational schema has been replaced with a direct link in the graph database.

There are two important dimensions of distinction. First is fine- versus coarse- grained metadata. Second is provenance versus workflow. Coarse-grained metadata describes how particular files come (or came) into being, or were (or could be) used to bring other files into being. Fine-grained metadata describes specific values recorded in social simulation outputs. To make the distinction concrete, suppose a simulation produces a CSV file. The data within

the CSV file are covered by fine-grained metadata, whilst the fact that the simulation produces the CSV file is coarse-grained. Turning to the other dimension, provenance metadata describes what actually happens (run W of simulation X produced output file Y), whilst workflow metadata describes what could happen (simulation X produces an output file of type Z). The distinctions are summarised in fig. 1.

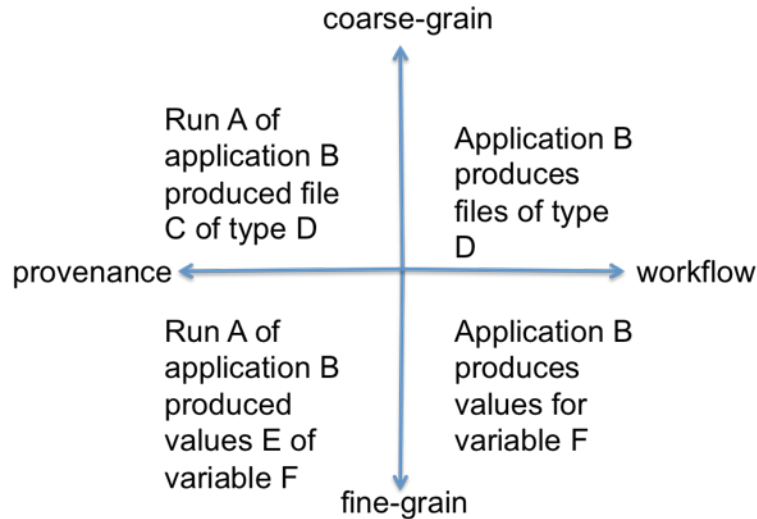


Figure 1: Dimensions of metadata

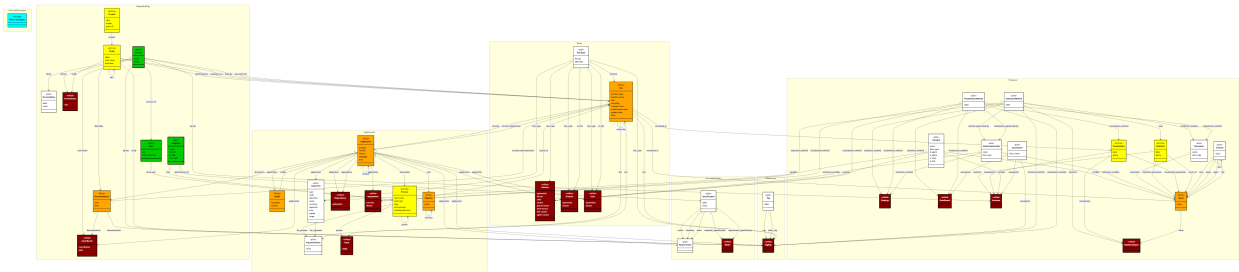


Figure 2: Metadata schema for MIRACLE. Boxes indicate tables, arcs represent relations, with 1 indicating a singular relations, "\*" indicating 'many' part of a many-to-one or many-to-many relationship. Tables are coloured in brown if they are breaking a many-to-many relationship (a reified relationship), with yellow, orange and green being used to denote specialisations from the PROV standard Activity, Entity and Agent classes respectively. Blue boxes show external databases/ontologies to which this one could be linked.

With the schema now having a large number of tables, it is better to consider it in parts. The following subgraphs of the schema (which may partially intersect) cover different topic areas of the whole:

- **Analysis:** Part of the fine grain pertaining only to analysis and visualisation.
- **External:** Links to external ontologies.
- **Fine-grain:** All fine-grain metadata.
- **Folksonomy:** Tables allowing tagging.
- **Project:** Metadata about projects.
- **Prov:** Tables capturing provenance metadata.
- **Services:** Tables pertaining to service-provision and matching requirements against specifications.
- **Workflow:** Tables relating to workflow.

The overview section now continues with an explanation of each subgraph in turn, with consideration given to the degree of user interaction and maintenance. The remainder of the document explains each table in detail.

### 3.1 Analysis

The subgraph for recording metadata about Analysis is shown in fig. 3. A `Variable` is metadata about the contents of files, and may have more than one `Value`. A `Value` may be visualised, and `Statistics` may be computed using them. `Statistics` are computed using `StatisticalMethods`, and `Visualisations` constructed using `VisualisationMethods`. `StatisticalMethods` produce `StatisticalVariables` as outputs, which are stored in the `Values` table as results-of `Statistics`. `Statistics` and `Visualisations` are thus conceived as a `PROV::Activity` in the `PROV` vocabulary (Gao et al. 2020).

When a user of the system has identified some statistics or visualisations they want to be able to reuse in other studies or record provenance about, as part of recording the activity, if the statistics are not something already added by another user, they would make an entry in the `StatisticalMethods`: or `VisualisationMethods` tables. If known, the user would also record any `Assumptions` Entailment by the methods. This information is not a requirement, as the user may not have relevant expertise to assert that a particular computation involves an `Assumption`; however, this information can be added at any time by any user. The `Employs` table can also be filled with data recording when one `StatisticalMethod` or `VisualisationMethod` uses a `StatisticalVariable` as input.

`Assumptions` could be quite trivial – in the most simple case, for example, that a numeric `Variable` is assumed to be a cardinal (as opposed to ordinal or nominal) when computing the mean of its `Values`. For other statistics, `Assumptions` can record such things as whether the `Variable` is normally distributed, or has constant variance.

Once an `Assumption` has been stated as `Entailment` by a `StatisticalMethod` or `VisualisationMethod`, it can be automatically inferred that `Persons` have made `Assumptions` about `Variables`, where they have done `Statistics` or `Visualisations` that use the `Methods` and `Values` of those `Variables` that are returned by the queries used to get the raw data on which the activities operated. The query is stored, along with the date made, in order to avoid populating large relational tables recording the `Visualisations` and `Statistics` that used `Values` as input data. Note that `Values` may also be used to store `Parameters` and `StatisticalVariables` that act to configure `Visualisations` and `Statistics`. These are captured using the `visualisation-parameter` and `statistical-parameter` relations, and the `StatisticalInput` table.

`Visualisations` are automatically inferred when the `Box` they are contained-in has a `BoxType` with a `VisualisationMethod` in it. The `Values` visualised (recorded in the `VisualisationValue` table) can be (possibly somewhat dubiously) inferred from the `Input` to the `Application` that ran the `Process` that created the `Box`, until such time as statistical tools support logging this information in sufficient detail.

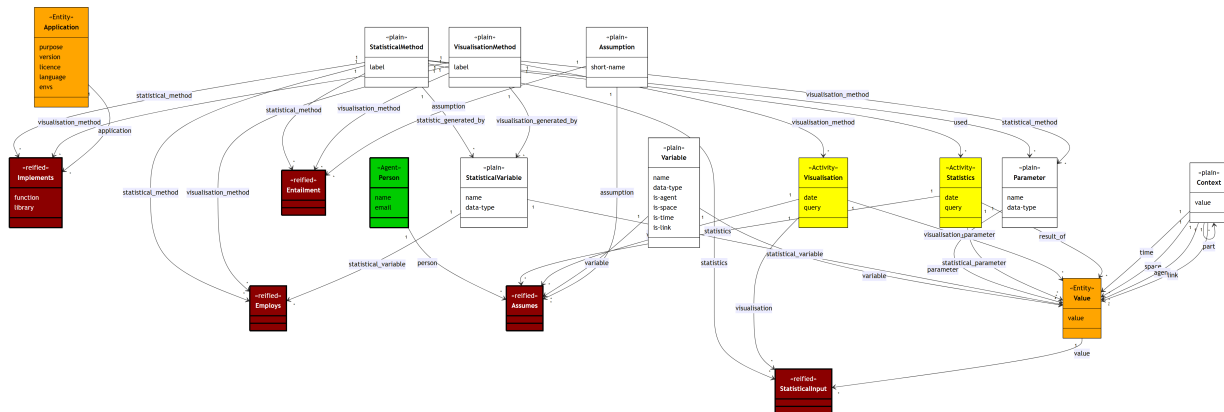


Figure 3: The analysis subgraph of the schema

## 3.2 External ontologies

Though not explicitly noted in the tables, with the exception of OpenABM (with which this schema had to integrate for the MIRACLE project), various external ontologies and schemas are relevant, and could be linked to if required. There are various ways such links could be manifested (relational tables for example, implementing each of the blue dashed lines in fig. 4), and those suggested are by no means exhaustive. Neither is it necessarily the case that any specific ontology or external schema is proposed or adopted. The following gives examples of specific external links:

- Bib – connection to a bibliographic database (e.g. bibsonomy). Note that the Documentation table is intended to be quite generic, and include journal and conference articles as well as reports and code documentation.
- Geo – links to ontologies or databases containing geographical or spatial concepts, such as GeoSparql and WGS84. The idea here is that we may want to link various table entries to external geographical databases, for example, to say that a Study pertained to a particular region, or that a Box is a GIS file.
- OpenABM – links to the CoMSES-Net archive of agent-based models, and is essential in order to identify which model these metadata are describing the output analysis of. (Although in principle, the system described could be applied to any stage of modelling, including setting up the model files, running the model and analysing the output, the focus of the MIRACLE project is specifically on the output analysis.)
- Services – links to vocabularies describing the requirements of applications and capabilities of service-providers. Where WSDL and OWL-S were once the natural reference points here, most service description in practice has since moved to REST-style APIs described using OpenAPI (formerly Swagger), which is now the de facto standard for machine-readable service interfaces. OpenAPI lacks OWL-S’s semantic-reasoning ambitions, but its wide tooling support and adoption make it a more practical link for automatic discovery of applications and services, including their input and output specifications.
- SocialWeb – we may want to allow people to link to social web tools such as ResearchGate, LinkedIn, Facebook and Twitter. The FOAF ontology also has attributes that we can draw on, and includes vocabulary for modelling social web links.
- Workflow – workflow-related ontologies. Standards in this space remain relatively immature. The Common Workflow Language (CWL) has emerged as the most widely adopted tool-agnostic specification for describing computational workflows, and is a natural candidate for linking here. Other approaches to modelling workflows, such as Petri-Nets and various business process modelling ontologies, remain potentially relevant, though PROV-O may be a more durable choice than domain-specific alternatives for capturing workflow provenance in particular (Herschel et al. 2017).
- GitHub/Zenodo – links to source-code repositories and their citable archived releases. GitHub is the natural place to link an Application’s container (in our terminology Box) to its version-controlled source code, and to the history of commits that produced it. However, since a GitHub repository is mutable and its URL is not guaranteed to remain stable, Zenodo’s GitHub integration – which mints a DOI for an immutable snapshot of a repository whenever a release is tagged – provides the more durable, citable record better suited to linking from the Documentation table, and is increasingly the standard route by which research software is formally cited in publications.

In terms of maintenance, provision for making the links to external databases would require users to supply the relevant information, except where specifically supported databases provided an API allowing us to query them for possibly relevant data to link to.

## 3.3 Fine grain

The fine grain metadata is information about the contents of files, including visualisations, and values of variables (fig. 5). Much of this has been covered already in the Analysis sec. 3.1 section above, but here we show how Boxes are disaggregated from a provenance perspective through saying that Values and Visualisations are contained-in them, and from a workflow perspective through saying that BoxTypes have Variables and VisualisationMethods as their content. The provenance side can be populated automatically, but the workflow metadata would need to be provided by the user when describing an Application they were making available to the system: specifically, the

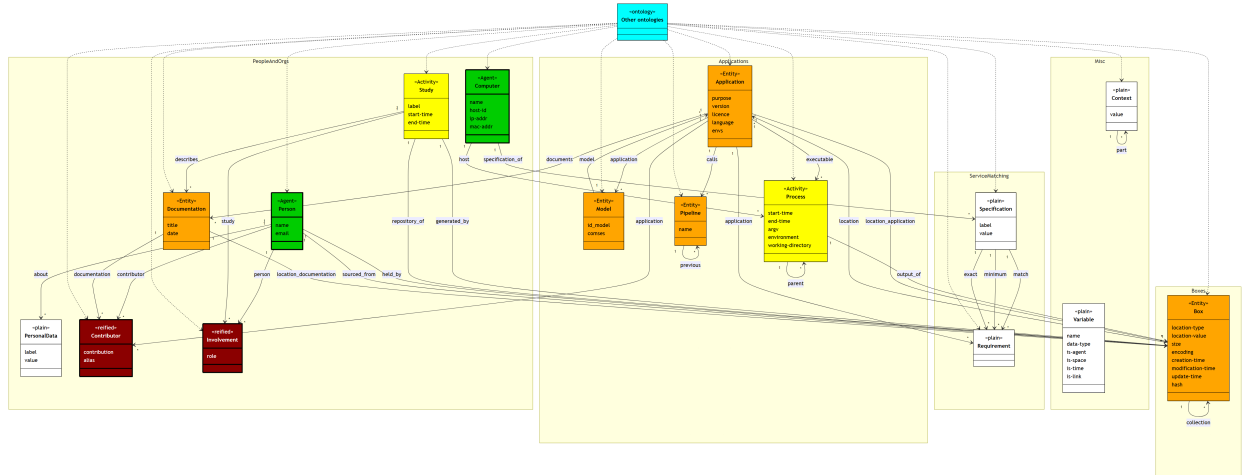


Figure 4: Subgraph showing possible tables that external ontologies could link to, and some relationships between them.

user will need to describe the inputs and outputs of each application, and for each associated BoxType, provide details of Variables and VisualisationMethods given as content. For some file formats, autodetection may assist with populating the Variable table – e.g. given an example of a CSV file used by or generated by an application, a header row could be used to suggest names for Variables they supply.

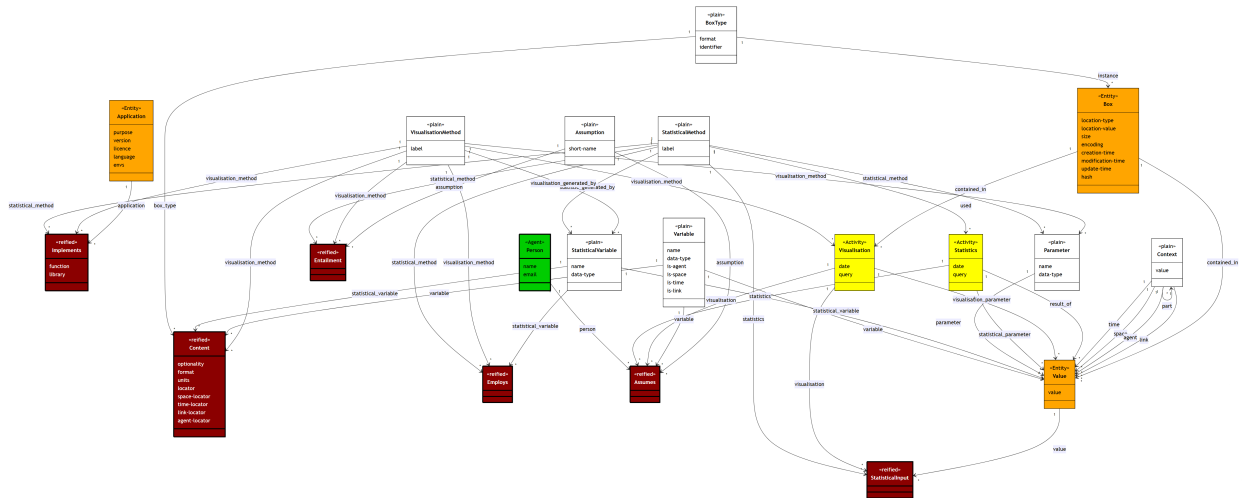


Figure 5: Subgraph showing the fine grain metadata tables

### 3.4 Folksonomy

Folksonomies are informal ontologies developed by a user communities, with tagging being a pretty much standard way in which this is achieved. It is not proposed to enrich that here. In fig. 6, the tables thought most likely to be of interest for users to tag are depicted, though this needn't be exhaustive. Each of the arcs from the Tag table would need a relational table to break the jmany-many relationship between tags and the concepts they are applied to.

### 3.5 Project

Project metadata (fig. 7) is largely for users to enter, and previously may have been regarded as unduly onerous (Edwards et al. 2014). However, it tells an important part of the story of a model from a Type 1 provenance (Pignotti

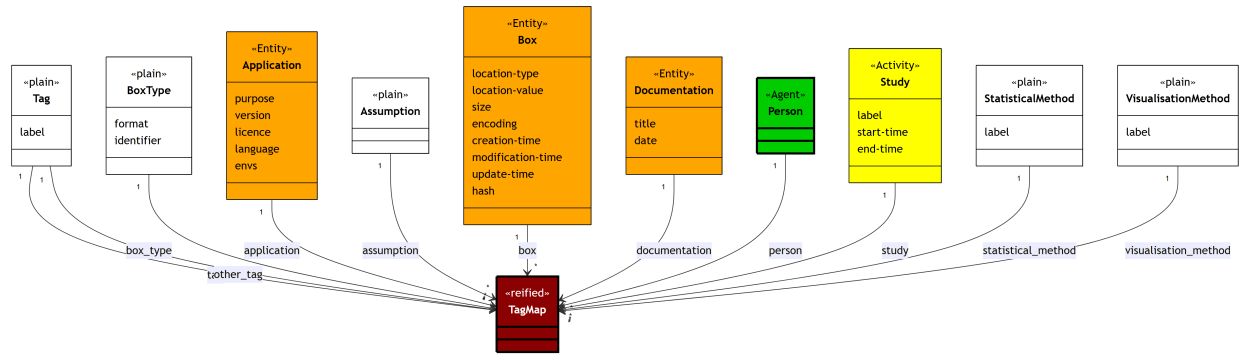


Figure 6: Folksonomy subgraph. Each of the arcs would itself be a reified table showing the application of Tags to tables.

et al. 2013) perspective, and with the advent of large language models, much of this metadata might be automatically generated. It is provided to facilitate users in understanding how simulation output data relates to publications (which would appear in the Documentation table), and specific pieces of work (Study table). The Study might be the most important table for users to complete, as this, by its association with Boxes allows collection of simulation outputs in useful groups.

### 3.6 Provenance

Provenance (fig. 8) is very much at the heart of the system, with an important role in recording how results from a model presented in journal articles are produced (Oliveira et al. 2018). Most of the tables here are autopopulated, as particularly for large scale analyses, user entry is an unrealistic goal. The system therefore needs to act as a wrapper around the scripts and tools (Applications) used to process the raw output from the simulation (which is the starting point of the MIRACLE project (Jin et al. 2017)) into the result used as part of an article. At the coarse grain, the central activity is the Process, which is a record of all relevant information of a process that ran on a Computer. This includes anything needed to replicate that Process under the same circumstances: the input files it used, command-line arguments, environment variables (essentially, anything that might affect its behaviour), and the output files it generated, the importance of which is illustrated by the challenges encountered when replicating the analysis of outputs from a social simulation of biodiversity incentivisation (Polhill et al. 2017).

The need for cross-platform support means that it is difficult to draw on standards for these. Different operating systems, and even different commands within the same operating systems, use different conventions for processing input on the command line, and input and output redirection. The Argument table attempts to record all relevant details about command-line arguments, allowing ArgumentValues to be inferred from a specific command-line instruction activating a Process. This is important in avoiding the need for users to supply too much information each time they run an Application.

Batch-file processing is assumed at the coarse grain; where processes involve user interaction that might affect the behaviour, automatic capture of provenance data would be extremely challenging. Whilst GUIs might be used during exploratory analysis of simulation output data, ultimately for reproducibility of results and keeping records of activities done (Pritchard and Wicenc 2025), these must ultimately be manifested as scripts that can be run in batch mode, once a decision is taken that a particular analysis step needs to be recorded as an application. Applications that only support GUI interaction can therefore not be supported (and arguably should be derided as virtually useless for any scientific endeavour – assuming repeatability is a desirable attribute of that work).

At the fine-grain, Statistics and Visualisations are the central activities. Again, although GUI interaction cannot be supported, command-line interaction potentially could. Parameters and StatisticalInput would need to be recorded, and the raw data on which the activities operate captured somehow in a query allowing the same set of data to be operated on. Recording the date at which the query was made is therefore important, especially if Values of Variables are subsequently populated that might affect the ability of future analyses to reuse the same data. A short-cut might be to store the date a Value was generated in the Values table, but since this table is expected to be

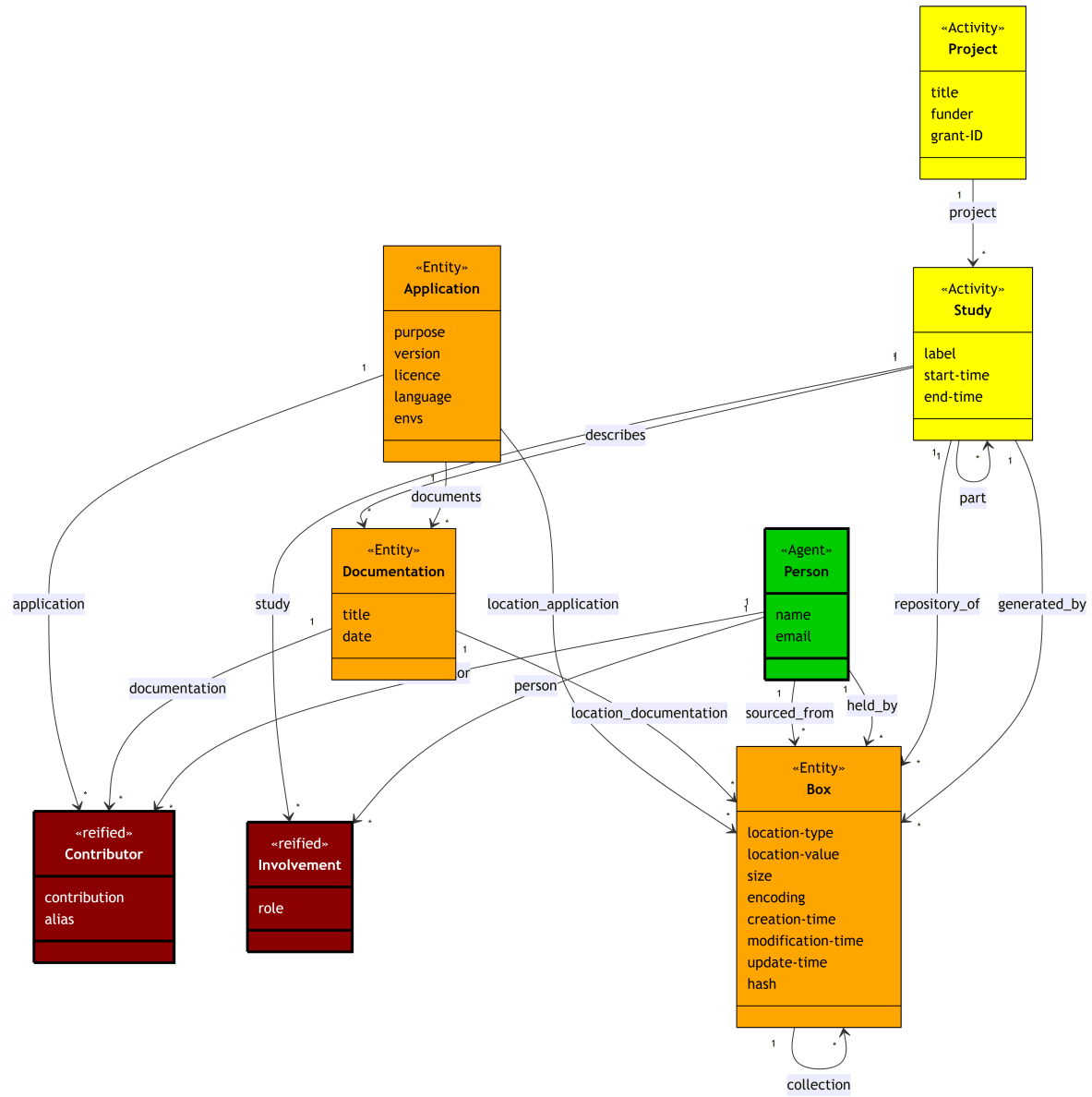


Figure 7: Subgraph capturing metadata about projects



'virtual' (in that a supporting system would gather relevant data from the raw data files), the date can be captured from the dates of the Processes generating the Boxes in which the Values appear, or the date at which the Statistics were computed if the Value is a StatisticalInput.

### 3.7 Services

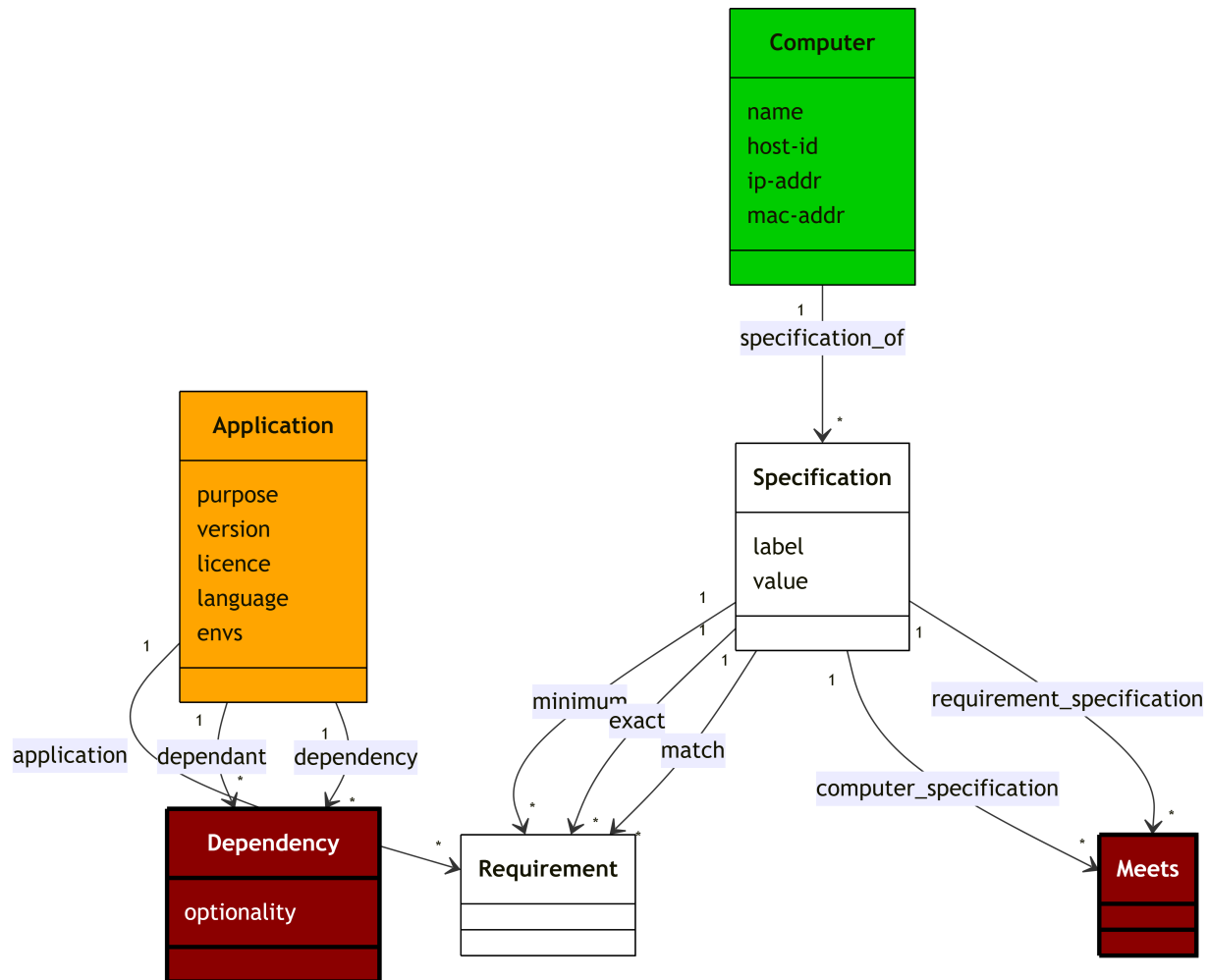


Figure 9: Services subgraph

The services part of the graph depicted in fig. 9 provides a simple model intended to be used to determine whether an Application can be run on a Computer. This involves checking the Requirement Specifications of the Application (and recursively of any Dependencies) against the Specifications of a Computer. Requirements can be specified in one of three ways – for numeric Requirement Specifications, a 'minimum' Requirement must be exceeded by the Specification of the Computer (e.g. for RAM). For all types of Requirement Specifications, an 'exact' Specification must be equal (an example might be the OS – if the Application has very specific demands), whilst a 'match' Specification is a regular expression that the Specification of the Computer must match. The results are stored in the Meets reified relationship (or as a direct link between the two if this a graph database).

A more sophisticated implementation would wrap each Application in a web service. This would also be more secure if responsibility for providing Applications as web services was in the hands of their developers – users of the framework would then not be uploading Applications to their own system, but instead sending data for

processing by other servers, and capturing metadata about these interactions. The services architecture could then draw much more heavily on web services ontologies, for example OWL-S (Martin et al. 2007) and WSDL (Christensen et al. 2001).

### 3.8 Workflow

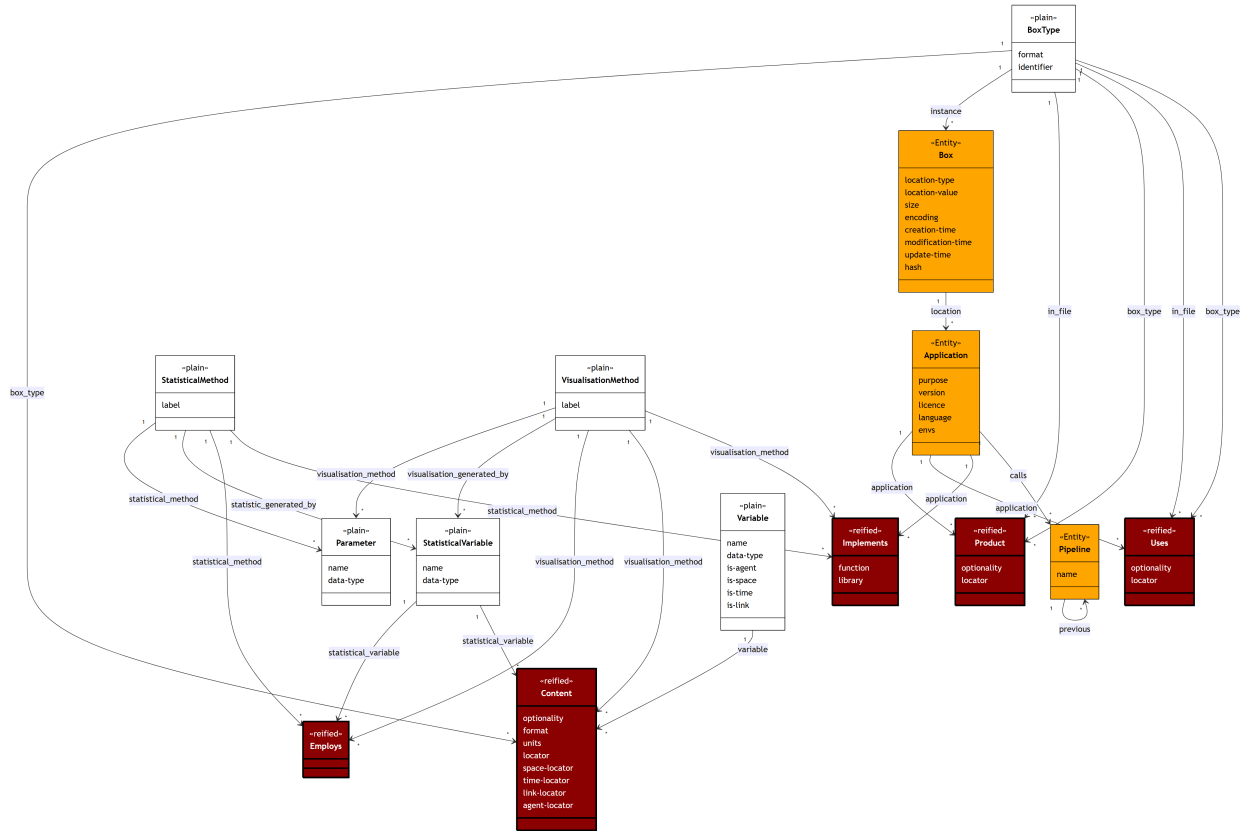


Figure 10: Workflow subgraph

The standard workflow model (fig. 10) provides prescriptive ‘recipes’ for undertaking specific procedures (Atkinson et al. 2017). Though we are interested in that here, and have provided a rudimentary sequence-based Pipeline table to handle that, we are also interested in providing support for exploring what ‘could be done’ given a current set of circumstances, and ‘what needs to be done’ before running an Application. For example, suppose we are interested in getting a Value for a particular Variable. Then we can see which BoxTypes have that Variable in their Content, and which Applications have the BoxType as a Product. If the BoxTypes those Applications Use have no instances, then we can find Applications that have the Uses BoxTypes as their Product, and so on. We can also search for Pipelines that have the required Product by exploring the Products of the last element of each Pipeline’s list-based structure, and checking the Uses of the first element.

This way, it should be possible, given a set of raw simulation outputs, to search for sequences of Applications to apply that generate a particular desired Visualisation(Method). Using the provenance infrastructure, it would also be possible to explore how other users have chosen to do it in the past.

## 4 Detailed specifications

All tables have uniquely specified ID fields as primary keys, unless they are associative tables (or direct links in the case of a graph database). All tables will have a name field. This is free form text and not always present. It identifies a relation, however it is not guaranteed to be unique. It is there to help primarily in readability when trying to extract information from this database and may contain a human readable label for the relation. For instance, for the definition of an argument, this would be the argument name. All tables will have a description field (dc:description). This field has been included to allow the introduction of free form documentary text. It is strongly advised that should the opportunity arise, then such text is supplied. The authors recommend this practice as it forms a useful means of documenting job control code in its own right, used to run these large scale models. Each table also has a creator/modifier field, and a corresponding created and modified date time field. Although most databases record this kind of metadata, we have made it explicit and expect these fields to be populated automatically. This is a deliberate attempt to make this interface database software agnostic. Note that date fields should be stored as strings conforming to the ISO 8601 date format, not in the native format of the database. This is recommended best practice according to the Dublin Core Metadata Initiative.<sup>[1]</sup> The remaining attribute that is always included in each relation is the "about" column. This we have pinched directly from RDF and uniquely identifies a triple in RDF (Hartig and Zhao 2010). Although currently optional, we intend this to uniquely identify the relationship within our database. The form of this is normally some kind of IRI. This will allow inward referencing of the provenance and metadata resource from other standardised resource software that recognizes IRIs.

### 4.0.1 Attributes

| Attribute   | Description                                                                                                   | Standards             | Validation                                                              | Automation                                                                                             |
|-------------|---------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| name        | A documentary name for the relation or entity                                                                 | None                  | None. May or may not be present                                         | None                                                                                                   |
| description | Short text to use to summarise what the argument is                                                           | None                  | String                                                                  | None                                                                                                   |
| about       | A unique resource identifier. This field will allow inward linking from any external resource that allow IRIs | RFC 3987 and RFC 4622 | IRI                                                                     | None                                                                                                   |
| creator     | Who created this particular set of attributes making up this relation                                         | dc:creator            | String                                                                  | Should be done by the agency that has implemented the SSREPI interface. This will be the computer user |
| created     | When this document was created                                                                                | dc:created, ISO8601   | logical date and time validation                                        | Should be done by the agency that has implemented the SSREPI interface. This will be the computer user |
| modifier    | Who created this particular set of attributes making up this relation                                         | dc:creator            | String                                                                  | Should be done by the agency that has implemented the SSREPI interface. This will be the computer user |
| modified    | When this document was created                                                                                | dc:modified, ISO8601  | Logical date and time validation. Must come after the date of creation. | Should be done by the agency that has implemented the SSREPI interface. This will be the computer user |

### 4.1 Annotation

This is not shown in the above diagrams. This is database entry that can annotate any relation in this database. This can be used to flag certain properties, or record information about a given set of rows. Such properties will be recorded in a JSON format for the purposes validation.

## 4.2 Standards

JSON for the annotation values. SQL for row restriction.

## 4.3 Automation

There is an entry which documents a database query (if suitable) that will document the rows affected. It is up to the standard implementor whether such queries will be allowed to be performed automatically. We are recommending only statements that refer to the Box table should be allowed to stop the possibility of SQL injections.

### 4.3.1 Attributes

| Attribute     | Description                                                                                                                                                                  | Standards | Validation     | Automation                               |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|----------------|------------------------------------------|
| id_annotation | Unique key                                                                                                                                                                   | None      | Must be unique | Automated by the instantiating framework |
| json          | json describing the values of the records being annotated.                                                                                                                   | None      | Must be unique | Automated by the instantiating framework |
| sql           | The SQL describing the range of records affected. This could be a single rows from single or multiple tables. BEWARE OF SQL CODE INJECTION when considering using this field | None      | Must be unique | Automated by the instantiating framework |

### 4.3.2 Relationships

None.

## 5 Application

### 5.1 Description

An *Application* is something that can be run by the user to generate or analyse simulation output.

### 5.2 Standards

PROV:Entity. Note the potential confusion. An *Application* is something that has the potential to be an activity (in the PROV sense) in the form of a *Process*. However, PROV only deals with the past, not with potential. The *Application* is a file somewhere, and hence an entity.

### 5.3 Automation

Most of this table is expected to be populated by the user.

### 5.4 Attributes

| Field          | Property    | Value                                    |
|----------------|-------------|------------------------------------------|
| ID_APPLICATION | Type        | TEXT                                     |
|                | Description | Unique key                               |
|                | Standards   | None                                     |
|                | Validation  | Must be unique                           |
|                | Automation  | Automated by the instantiating framework |

| Field     | Property    | Value                                                                         |
|-----------|-------------|-------------------------------------------------------------------------------|
| PURPOSE   | Type        | TEXT                                                                          |
|           | Description | Description of the purpose of the application                                 |
|           | Standards   | ?                                                                             |
|           | Validation  | Free text                                                                     |
| VERSION   | Automation  | None                                                                          |
|           | Type        | TEXT                                                                          |
|           | Description | Version ID for the application                                                |
|           | Standards   | dc:?                                                                          |
| LICENCE   | Validation  | String (e.g. CHAR(32))                                                        |
|           | Automation  | None                                                                          |
|           | Type        | TEXT                                                                          |
|           | Description | Software licence for the application                                          |
| LANGUAGE  | Standards   | dc:license                                                                    |
|           | Validation  | Free text with the option to select from other entries                        |
|           | Automation  | None                                                                          |
|           | Type        | TEXT                                                                          |
| ENVS      | Description | Programming language the application was written in                           |
|           | Standards   | ?                                                                             |
|           | Validation  | Free text with the option to select from other entries                        |
|           | Automation  | None                                                                          |
| SEPARATOR | Type        | TEXT                                                                          |
|           | Description | The environment variables that should be recorded when the application is run |
|           | Standards   | None                                                                          |
|           | Validation  | List of strings                                                               |
|           | Automation  | None                                                                          |
|           | Type        | TEXT                                                                          |
|           | Description | Separator used when supplying arguments to the application                    |
|           | Standards   | None                                                                          |
|           | Validation  | String                                                                        |
|           | Automation  | None                                                                          |

## 5.5 Relationships

| Field    | Property    | Value                                                                                                                                                                 |
|----------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REVISION | Type        | TEXT                                                                                                                                                                  |
|          | Description | ID in Application table of Application this Application is a revision of, if any.                                                                                     |
|          | Standards   | PROV:wasDerivedFrom; dc:isVersionOf                                                                                                                                   |
|          | Validation  | Must be an ID of an Application                                                                                                                                       |
| MODEL    | Null        | If not a revision of another Application.                                                                                                                             |
|          | Automation  | None                                                                                                                                                                  |
|          | Type        | TEXT                                                                                                                                                                  |
|          | Description | ID in Model table this Application is, if it is a Model.                                                                                                              |
| LOCATION | Standards   | None                                                                                                                                                                  |
|          | Validation  | Must be an ID in Model.                                                                                                                                               |
|          | Null        | If this Application is not a model.                                                                                                                                   |
|          | Automation  | None                                                                                                                                                                  |
|          | Type        | TEXT                                                                                                                                                                  |
|          | Description | ID_BOX in the Box table to use to find this Application.                                                                                                              |
|          | Standards   | ?                                                                                                                                                                     |
|          | Validation  | Must be ID_BOX of a Box.                                                                                                                                              |
|          | Null        | If the search for a Box for this Application has not been done.                                                                                                       |
|          | Automation  | This should be populated automatically the first time the Application is requested on a host by finding the most local Box that references it, and storing that here. |

If the most local Box is not on the current host, then the Application should be downloaded to the current host, and a new Box created for the location.

If the Box ID is no longer present, then the search should be repeated in the Box table. |

## 6 Argument

### 6.1 Description

A command-line argument accepted by an `Application`. Commands vary hugely in how they parse arguments on the command line, and this table needs to make clear how to build a command line that the `Application` can use. To be clear, a command line is a string of text that is given to a shell (DOS, bash, etc.) to initiate a batch job.

### 6.2 Standards

POSIX.1-2008 and equivalently IEEE Std 1003.1-2008/Cor 1-2013 are normative for Unix environments, but cannot be assumed even for Unix environments. Cross-platform support is in any case needed.

### 6.3 Automation

When building up a command-line for an `Application` with arguments during invocation of a `Process`, the system should do the following:

- Let `M` be a map from integer to list of string.
- For each flag, check with the user to see whether it should be set or cleared; if set, add the flag name to `M`, using the order as the key, or `-1` if the order is null. Create an entry in `ArgumentValue` using 'true' or 'false' as the value according to whether or not the flag is set.
- For each option, check with the user to see whether it should be used, and if so, provide an appropriate argument or arguments in accordance with the arity. Build a string for the option including its name and arguments, bearing in mind the separator and `argsep` values. Add the resulting string to `M`.
- For each required argument, request a value or values from the user in accordance with the arity. Build a string accordingly and add to `M`.
- Let `K` be a sorted list of keys of `M`.
- Let `C` be a string.
- For each key, join its list of strings with space and concatenate to `C`.
- `C` contains the command-line argument string.

### 6.4 Attributes

| Field               | Property    | Value                                                                                                                                                                                                                                                                          |
|---------------------|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ID_ARGUMENT         | Type        | TEXT                                                                                                                                                                                                                                                                           |
|                     | Description | Unique key                                                                                                                                                                                                                                                                     |
|                     | Standards   | None                                                                                                                                                                                                                                                                           |
|                     | Validation  | Must be unique                                                                                                                                                                                                                                                                 |
|                     | Automation  | Automated by the instantiating framework                                                                                                                                                                                                                                       |
| TYPE                | Type        | TEXT                                                                                                                                                                                                                                                                           |
|                     | Description | The kind of command line argument this is. Required arguments must be provided, typically don't have a name, and are identified by their number. Options typically have a name, which if given stipulates some sort of value must be provided. Flags are options with arity 0. |
|                     | Standards   | None                                                                                                                                                                                                                                                                           |
|                     | Validation  | One of "required", "option", or "flag"                                                                                                                                                                                                                                         |
|                     | Automation  | None                                                                                                                                                                                                                                                                           |
| ORDER_VALUE         | Type        | INTEGER                                                                                                                                                                                                                                                                        |
|                     | Description | A number used to indicate any order in which this <code>Argument</code> should appear in relation to other <code>Arguments</code> the <code>Application</code> accepts.                                                                                                        |
|                     | Standards   | None                                                                                                                                                                                                                                                                           |
|                     | Validation  | Non-negative integer or null if the order is unimportant.                                                                                                                                                                                                                      |
|                     | Automation  | None                                                                                                                                                                                                                                                                           |
| ASSIGNMENT_OPERATOR | Type        | TEXT                                                                                                                                                                                                                                                                           |
|                     | Description | Separator to use between argument name and value.                                                                                                                                                                                                                              |
|                     | Standards   | None                                                                                                                                                                                                                                                                           |
|                     | Validation  | String                                                                                                                                                                                                                                                                         |
|                     | Automation  | None                                                                                                                                                                                                                                                                           |
| NAME                | Type        | TEXT                                                                                                                                                                                                                                                                           |
|                     | Description | The name of the argument, including any grammar to indicate on the command-line that it is an argument (such as <code>--</code> or <code>/</code> ).                                                                                                                           |
|                     | Standards   | None                                                                                                                                                                                                                                                                           |
|                     | Validation  | String                                                                                                                                                                                                                                                                         |
|                     | Automation  | None                                                                                                                                                                                                                                                                           |

| Field           | Property    | Value                                                                                                                                                                                                |
|-----------------|-------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| SEPARATOR       | Type        | TEXT                                                                                                                                                                                                 |
|                 | Description | This is the character that indicates the argument name (e.g. -input, -input, /input).                                                                                                                |
|                 | Standards   | None                                                                                                                                                                                                 |
|                 | Validation  | String                                                                                                                                                                                               |
| SHORT_NAME      | Automation  | None                                                                                                                                                                                                 |
|                 | Type        | TEXT                                                                                                                                                                                                 |
|                 | Description | The short name of the argument, including any grammar to indicate on the command-line that it is an argument (such as -- or - or /). For example in *nix system -h and -help are the same parameter. |
|                 | Standards   | None                                                                                                                                                                                                 |
| SHORT_SEPARATOR | Validation  | String                                                                                                                                                                                               |
|                 | Automation  | None                                                                                                                                                                                                 |
|                 | Type        | TEXT                                                                                                                                                                                                 |
|                 | Description | This is the character that indicates the argument name (e.g. -input, -input, /input). for example in *nix system -h and -help are the same parameter.                                                |
| DESCRIPTION     | Standards   | None                                                                                                                                                                                                 |
|                 | Validation  | String                                                                                                                                                                                               |
|                 | Automation  | None                                                                                                                                                                                                 |
|                 | Type        | TEXT                                                                                                                                                                                                 |
| ARITY           | Description | Short text to use to summarise what the argument is                                                                                                                                                  |
|                 | Standards   | None                                                                                                                                                                                                 |
|                 | Validation  | String                                                                                                                                                                                               |
|                 | Automation  | None                                                                                                                                                                                                 |
| ARGSEP          | Type        | TEXT                                                                                                                                                                                                 |
|                 | Description | Number of arguments expected. Can be integer, ?, + or *.                                                                                                                                             |
|                 | Standards   | None                                                                                                                                                                                                 |
|                 | Validation  | ?, + or *, or integer string. Constraints apply.                                                                                                                                                     |
| RANGE           | Automation  | None                                                                                                                                                                                                 |
|                 | Type        | TEXT                                                                                                                                                                                                 |
|                 | Description | Separator to use between arguments if arity > 1                                                                                                                                                      |
|                 | Standards   | None                                                                                                                                                                                                 |
|                 | Validation  | String                                                                                                                                                                                               |
|                 | Automation  | None                                                                                                                                                                                                 |
|                 | Type        | TEXT                                                                                                                                                                                                 |
|                 | Description | Description of the range of any value to be supplied by the user                                                                                                                                     |
|                 | Standards   | None                                                                                                                                                                                                 |
|                 | Validation  | String                                                                                                                                                                                               |
|                 | Automation  | None                                                                                                                                                                                                 |

## 6.5 Relationships

| Field       | Property    | Value                                                            |
|-------------|-------------|------------------------------------------------------------------|
| APPLICATION | Type        | TEXT                                                             |
|             | Description | The application this argument applies to.                        |
|             | Standards   | None                                                             |
|             | Validation  | ID in the Application table.                                     |
| VARIABLE    | Null        | Null if not an argument for an Application                       |
|             | Automation  | None                                                             |
|             | Type        | TEXT                                                             |
|             | Description | The variable this argument relates to.                           |
| BOX_TYPE    | Standards   | None                                                             |
|             | Validation  | ID in the Variables table.                                       |
|             | Null        | Null if not about a variable                                     |
|             | Automation  | None                                                             |
|             | Type        | TEXT                                                             |
|             | Description | The BoxType this argument might be, if it is a file or somesuch. |
|             | Standards   | None                                                             |
|             | Validation  | ID in the BoxTypes table.                                        |
|             | Null        | Null if this does not have a BoxType associated iwth it.         |
|             | Automation  | None                                                             |

## 7 ArgumentValue

### 7.1 Description

A value supplied for a command-line argument in a run of an Application.

### 7.2 Standards

None

### 7.3 Automation

Populated automatically when a Process is invoked.

### 7.4 Attributes

| Field        | Property    | Value                                                            |
|--------------|-------------|------------------------------------------------------------------|
| HAS_VALUE    | Type        | TEXT                                                             |
|              | Description | Value supplied, or true/false for flags                          |
|              | Standards   | None                                                             |
|              | Validation  | None                                                             |
|              | Automation  | Populated when the Process is invoked                            |
| FOR_PROCESS  | Type        | TEXT                                                             |
|              | Description | ID in Process table of the Process this ArgumentValue applies to |
|              | Standards   | None                                                             |
|              | Validation  | Must be an ID of a Process                                       |
|              | Null        | Not null                                                         |
| FOR_ARGUMENT | Automation  | Populated automatically when the Process is created              |
|              | Type        | TEXT                                                             |
|              | Description | ID in Argument table of the Argument this value is for           |
|              | Standards   | None                                                             |
|              | Validation  | Must be an ID of an Argument                                     |
| BOX          | Null        | Not null                                                         |
|              | Automation  | Populated automatically when the Process is created              |
|              | Type        | TEXT                                                             |
|              | Description | ID in Box table of the box this value can be for                 |
|              | Standards   | None                                                             |
|              | Validation  | Must be an ID of a Box                                           |
|              | Null        | Not null                                                         |
|              | Automation  | Populated automatically when the Process is created              |

## 8 Assumes

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 8.1 Standards

None

### 8.2 Automation

This relationship may be inferred automatically from the use of StatisticalMethod or VisualisationMethod.

### 8.3 Relationships

| Field         | Property    | Value                                                                     |
|---------------|-------------|---------------------------------------------------------------------------|
| PERSON        | Type        | TEXT                                                                      |
|               | Description | ID in Person table of the Person making the assumption                    |
|               | Standards   | None                                                                      |
|               | Validation  | Must be an ID of a Person                                                 |
|               | Null        | Not null                                                                  |
| STATISTICS    | Automation  | None                                                                      |
|               | Type        | TEXT                                                                      |
|               | Description | ID in Statistics table if assumption applies to a statistical computation |
|               | Standards   | None                                                                      |
|               | Validation  | Must be an ID of a Statistics                                             |
| VISUALISATION | Null        | Null if Visualisation is not null                                         |
|               | Automation  | None                                                                      |
|               | Type        | TEXT                                                                      |
|               | Description | ID in Visualisation table if assumption applies to a Visualisation        |
|               | Standards   | None                                                                      |
| VARIABLE      | Validation  | Must be an ID of a Visualisation                                          |
|               | Null        | Null if Statistics is not null                                            |
|               | Automation  | None                                                                      |
|               | Type        | TEXT                                                                      |
|               | Description | ID in Variable table of the variable to which the assumption applies      |
| ASSUMPTION    | Standards   | None                                                                      |
|               | Validation  | Must be an ID of a Variable                                               |
|               | Null        | Not null                                                                  |
|               | Automation  | None                                                                      |
|               | Type        | TEXT                                                                      |
|               | Description | ID in Assumption table of the Assumption being made                       |
|               | Standards   | None                                                                      |
|               | Validation  | Must be an ID of an Assumption                                            |
|               | Null        | Not null                                                                  |
|               | Automation  | None                                                                      |

## 9 Assumption

### 9.1 Description

An **Assumption** is a condition applied to a **Variable** for its proper application to a **Statistic**. (That **Statistic** being realised as an aggregation of the **Variable** to which the **Assumption** applies.) A **Person** makes an **Assumption** about a **Variable** (in the **Assumes** table), explicitly or implicitly, every time they compute the **Statistic** on it.

### 9.2 Standards

None

### 9.3 Automation

None

### 9.4 Attributes

| Field         | Property    | Value                         |
|---------------|-------------|-------------------------------|
| ID_ASSUMPTION | Type        | TEXT                          |
|               | Description | Short name for the assumption |
|               | Standards   | None                          |
|               | Validation  | Must be unique                |
|               | Automation  | None                          |

## 10 Box

### 10.1 Description

A Box is any data container (file, database, URI, etc.) used to store or reference data within the system.

### 10.2 Standards

PROV:Entity

### 10.3 Automation

A Box should be created automatically whenever data is accessed or generated by a Process. Metadata such as size, encoding, timestamps, and hash may be populated automatically using system tools (e.g. OS calls, HTTP headers, checksum utilities).

### 10.4 Attributes

| Field             | Property    | Value                                                                    |
|-------------------|-------------|--------------------------------------------------------------------------|
| ID_BOX            | Type        | TEXT                                                                     |
|                   | Description | Unique identifier for the Box                                            |
|                   | Standards   | None                                                                     |
|                   | Validation  | Primary key                                                              |
| LOCATION_TYPE     | Automation  | Generated automatically                                                  |
|                   | Type        | TEXT                                                                     |
|                   | Description | Type of resource the Box refers to (e.g. file, URI, database)            |
|                   | Standards   | ?                                                                        |
| LOCATION_VALUE    | Validation  | String describing access type                                            |
|                   | Automation  | Requires appropriate software depending on type                          |
|                   | Type        | TEXT                                                                     |
|                   | Description | The means of accessing the resource (path, URI, connection string, etc.) |
| SIZE              | Standards   | URI/IRI                                                                  |
|                   | Validation  | Depends on LOCATION-TYPE                                                 |
|                   | Automation  | Parsed and used by system handlers                                       |
|                   | Type        | INTEGER                                                                  |
| ENCODING          | Description | Size of the resource in bytes                                            |
|                   | Standards   | ISO/IEC 80000-13                                                         |
|                   | Validation  | Non-negative integer or null                                             |
|                   | Automation  | Retrieved via OS or HTTP headers                                         |
| CREATION_TIME     | Type        | TEXT                                                                     |
|                   | Description | Encoding of the resource                                                 |
|                   | Standards   | MIME                                                                     |
|                   | Validation  | Valid MIME type or null                                                  |
| MODIFICATION_TIME | Automation  | Detected using system tools (e.g. file -l, HTTP headers)                 |
|                   | Type        | TEXT                                                                     |
|                   | Description | Time the resource was created                                            |
|                   | Standards   | ISO8601                                                                  |
| UPDATE_TIME       | Validation  | Datetime string                                                          |
|                   | Automation  | Retrieved via OS stat or equivalent                                      |
|                   | Type        | TEXT                                                                     |
|                   | Description | Last modification time of the resource                                   |
| HASH              | Standards   | ISO8601                                                                  |
|                   | Validation  | Datetime string                                                          |
|                   | Automation  | Retrieved via OS stat or HTTP headers                                    |
|                   | Type        | TEXT                                                                     |
|                   | Description | Time the metadata for this Box was last updated                          |
|                   | Standards   | ISO8601                                                                  |
|                   | Validation  | Datetime string                                                          |
|                   | Automation  | System maintained                                                        |
|                   | Type        | TEXT                                                                     |
|                   | Description | Hash of the resource content                                             |

| Field    | Property    | Value                                                  |
|----------|-------------|--------------------------------------------------------|
| INSTANCE | Standards   | None                                                   |
|          | Validation  | Format: algorithm:encoding:value                       |
|          | Automation  | Generated using hashing tools (e.g. md5sum, sha256sum) |
|          | Type        | TEXT                                                   |
|          | Description | ID in BoxType table describing the type of Box         |
|          | Standards   | None                                                   |
|          | Validation  | Must be an ID of a BoxType                             |
|          | Null        | Not null                                               |
|          | Automation  | None                                                   |

## 10.5 Relationships

| Field                  | Property    | Value                                                         |
|------------------------|-------------|---------------------------------------------------------------|
| LOCATION_APPLICATION   | Type        | TEXT                                                          |
|                        | Description | ID in Application table if this Box refers to an Application  |
|                        | Standards   | None                                                          |
|                        | Validation  | Must be an ID of an Application.                              |
|                        | Null        | Null if not applicable                                        |
|                        | Automation  | None                                                          |
| LOCATION_DOCUMENTATION | Type        | TEXT                                                          |
|                        | Description | ID in Documentation table if this Box refers to Documentation |
|                        | Standards   | None                                                          |
|                        | Validation  | Must be an ID of Documentation                                |
|                        | Null        | Null if not applicable                                        |
|                        | Automation  | None                                                          |
| GENERATED_BY           | Type        | TEXT                                                          |
|                        | Description | ID in Study table of the Study that generated this Box        |
|                        | Standards   | PROV:wasGeneratedBy                                           |
|                        | Validation  | Must be an ID of a Study                                      |
|                        | Null        | Null if not generated by a Study                              |
|                        | Automation  | Set when Process produces output                              |
| REPOSITORY_OF          | Type        | TEXT                                                          |
|                        | Description | ID in Study table of Study this Box is a repository for       |
|                        | Standards   | None                                                          |
|                        | Validation  | Must be an ID of a Study                                      |
|                        | Null        | Optional                                                      |
|                        | Automation  | None                                                          |
| HELD_BY                | Type        | TEXT                                                          |
|                        | Description | ID in Person table of person holding the Box                  |
|                        | Standards   | PROV:wasAttributedTo                                          |
|                        | Validation  | Must be an ID of a Person                                     |
|                        | Null        | Optional                                                      |
|                        | Automation  | None                                                          |
| SOURCED_FROM           | Type        | TEXT                                                          |
|                        | Description | ID in Person table of source of the Box                       |
|                        | Standards   | PROV:wasAttributedTo                                          |
|                        | Validation  | Must be an ID of a Person                                     |
|                        | Null        | Optional                                                      |
|                        | Automation  | None                                                          |
| OUTPUT_OF              | Type        | TEXT                                                          |
|                        | Description | ID in Process table of the Process that produced this Box     |
|                        | Standards   | PROV:wasGeneratedBy                                           |
|                        | Validation  | Must be an ID of a Process                                    |
|                        | Null        | Null if not produced by a Process                             |
|                        | Automation  | Automatically set during Process execution                    |
| COLLECTION             | Type        | TEXT                                                          |
|                        | Description | ID in Box table if this Box is part of another Box            |
|                        | Standards   | PROV:hadMember                                                |
|                        | Validation  | Must be an ID of a Box                                        |
|                        | Null        | Null if not part of a collection                              |
|                        | Automation  | None                                                          |

## 11 BoxType

### 11.1 Description

A `BoxType` defines the format and identification rules for Boxes, describing how their contents should be interpreted.

### 11.2 Standards

None

### 11.3 Automation

`BoxTypes` are expected to be defined by the user; identification rules may be applied automatically when inspecting Box contents.

### 11.4 Attributes

| Field       | Property    | Value                                                                                                          |
|-------------|-------------|----------------------------------------------------------------------------------------------------------------|
| ID_BOX_TYPE | Type        | TEXT                                                                                                           |
|             | Description | Unique identifier for the <code>BoxType</code>                                                                 |
|             | Standards   | None                                                                                                           |
|             | Validation  | Must be unique                                                                                                 |
| FORMAT      | Automation  | Automated by the instantiating framework                                                                       |
|             | Type        | TEXT                                                                                                           |
|             | Description | Format of the Box contents                                                                                     |
|             | Standards   | MIME                                                                                                           |
| IDENTIFIER  | Validation  | Valid MIME type                                                                                                |
|             | Automation  | None                                                                                                           |
|             | Type        | TEXT                                                                                                           |
|             | Description | Rule used to identify whether a Box conforms to this <code>BoxType</code> (e.g. magic bytes, filename pattern) |
|             | Standards   | None                                                                                                           |
|             | Validation  | Structured rule string                                                                                         |
|             | Automation  | None                                                                                                           |

## 12 Computer

### 12.1 Description

A `Computer` represents a machine on which a `Process` is executed.

### 12.2 Standards

PROV:Agent

### 12.3 Automation

Most values are expected to be automatically obtained from the operating system and network interfaces.

### 12.4 Attributes

| Field       | Property    | Value      |
|-------------|-------------|------------|
| ID_COMPUTER | Type        | TEXT       |
|             | Description | Unique key |
|             | Standards   | None       |

| Field       | Property    | Value                                    |
|-------------|-------------|------------------------------------------|
| HOST_ID     | Validation  | Must be unique                           |
|             | Automation  | Automated by the instantiating framework |
|             | Type        | TEXT                                     |
|             | Description | Identifier for the host machine          |
| IP_ADDRESS  | Standards   | ?                                        |
|             | Validation  | String                                   |
|             | Automation  | Retrieved from the operating system      |
|             | Type        | TEXT                                     |
| MAC_ADDRESS | Description | IP address of the machine                |
|             | Standards   | ?                                        |
|             | Validation  | Valid IP address string                  |
|             | Automation  | Retrieved from the network interface     |
|             | Type        | TEXT                                     |
|             | Description | MAC address of the machine               |
|             | Standards   | ?                                        |
|             | Validation  | Valid MAC address string                 |
|             | Automation  | Retrieved from the network interface     |

## 13 Content

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 13.1 Description

The `Content` table describes how a `Variable` (or `StatisticalVariable`) is located within a `Box`.

### 13.2 Standards

None

### 13.3 Automation

None

### 13.4 Attributes

| Field        | Property    | Value                                                                                                     |
|--------------|-------------|-----------------------------------------------------------------------------------------------------------|
| OPTIONALITY  | Type        | TEXT                                                                                                      |
|              | Description | Whether the <code>Variable</code> always appears or only appears depending on some condition              |
|              | Standards   | None                                                                                                      |
|              | Validation  | One of 'always' or 'depends'                                                                              |
| LOCATOR      | Automation  | None                                                                                                      |
|              | Type        | TEXT                                                                                                      |
|              | Description | How to locate the <code>Variable</code> value within the <code>Box</code> (e.g. row:X, column:Y, field:Z) |
|              | Validation  | Formatted rule string                                                                                     |
| TIME_LOCATOR | Automation  | None                                                                                                      |
|              | Type        | TEXT                                                                                                      |
|              | Description | How to locate the temporal context (e.g. timestamps)                                                      |
|              | Standards   | None                                                                                                      |
| LINK_LOCATOR | Validation  | Formatted rule string                                                                                     |
|              | Automation  | None                                                                                                      |
|              | Type        | TEXT                                                                                                      |
|              | Description | How to locate link identifiers (for relational data)                                                      |
|              | Standards   | None                                                                                                      |
|              | Validation  | Formatted rule string                                                                                     |
|              | Automation  | None                                                                                                      |

| Field                | Property    | Value                                                                                                                  |
|----------------------|-------------|------------------------------------------------------------------------------------------------------------------------|
| AGENT_LOCATOR        | Type        | TEXT                                                                                                                   |
|                      | Description | How to locate agent identifiers                                                                                        |
|                      | Standards   | None                                                                                                                   |
|                      | Validation  | Formatted rule string                                                                                                  |
|                      | Automation  | None                                                                                                                   |
| STATISTICAL_VARIABLE | Type        | TEXT                                                                                                                   |
|                      | Description | ID in <code>StatisticalVariable</code> table if this <code>Content</code> refers to a <code>StatisticalVariable</code> |
|                      | Standards   | None                                                                                                                   |
|                      | Validation  | Must be an ID of a <code>StatisticalVariable</code>                                                                    |
|                      | Null        | Null if <code>VARIABLE</code> is not null                                                                              |
|                      | Automation  | None                                                                                                                   |

## 13.5 Relationships

| Field                | Property    | Value                                                                                               |
|----------------------|-------------|-----------------------------------------------------------------------------------------------------|
| BOX_TYPE             | Type        | TEXT                                                                                                |
|                      | Description | ID in <code>BoxType</code> table of the <code>BoxType</code> this <code>Content</code> applies to   |
|                      | Standards   | None                                                                                                |
|                      | Validation  | Must be an ID of a <code>BoxType</code>                                                             |
|                      | Null        | Not null                                                                                            |
|                      | Automation  | None                                                                                                |
| VARIABLE             | Type        | TEXT                                                                                                |
|                      | Description | ID in <code>Variable</code> table if this <code>Content</code> refers to a <code>Variable</code>    |
|                      | Standards   | None                                                                                                |
|                      | Validation  | Must be an ID of a <code>Variable</code>                                                            |
|                      | Null        | Null if <code>STATISTICAL_VARIABLE</code> is not null                                               |
|                      | Automation  | None                                                                                                |
| VISUALISATION_METHOD | Type        | TEXT                                                                                                |
|                      | Description | ID in <code>VisualisationMethod</code> table if this <code>Content</code> is used for visualisation |
|                      | Standards   | None                                                                                                |
|                      | Validation  | Must be an ID of a <code>VisualisationMethod</code>                                                 |
|                      | Null        | Optional                                                                                            |
|                      | Automation  | None                                                                                                |

## 14 Context

### 14.1 Description

The `Context` table provides contextual values such as time, space, agent, or link, which can be associated with `Values`.

### 14.2 Standards

None

### 14.3 Automation

`Context` entries may be created automatically when extracting or interpreting data from `Boxes`.

### 14.4 Attributes

| Field      | Property    | Value                                                              |
|------------|-------------|--------------------------------------------------------------------|
| ID_CONTEXT | Type        | TEXT                                                               |
|            | Description | Unique identifier for the Context                                  |
|            | Standards   | None                                                               |
|            | Validation  | Must be unique                                                     |
| VALUE      | Automation  | Automated by the instantiating framework                           |
|            | Type        | TEXT                                                               |
|            | Description | Value of the context (e.g. time, space, agent, or link identifier) |
|            | Standards   | None                                                               |
|            | Validation  | String                                                             |
|            | Automation  | Derived from Content locators or data extraction                   |

## 14.5 Relationships

| Field | Property    | Value                                                                       |
|-------|-------------|-----------------------------------------------------------------------------|
| PART  | Type        | TEXT                                                                        |
|       | Description | ID in Context table indicating that this Context is part of another Context |
|       | Standards   | None                                                                        |
|       | Validation  | Must be an ID of a Context                                                  |
|       | Null        | Null if this Context is not part of another                                 |
|       | Automation  | None                                                                        |

## 15 Contributor

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 15.1 Description

The Contributor table records people who have contributed to Applications or Documentation, including how they are credited.

### 15.2 Standards

dc:contributor

### 15.3 Automation

None

### 15.4 Attributes

| Field        | Property    | Value                                                                                     |
|--------------|-------------|-------------------------------------------------------------------------------------------|
| CONTRIBUTION | Type        | TEXT                                                                                      |
|              | Description | Nature of the contribution made by the person                                             |
|              | Standards   | dc:contributor                                                                            |
|              | Validation  | String describing contribution type                                                       |
|              | Automation  | None                                                                                      |
| ALIAS        | Type        | TEXT                                                                                      |
|              | Description | Name or alias used for the contributor in the context of the Application or Documentation |
|              | Standards   | None                                                                                      |
|              | Validation  | String                                                                                    |
|              | Automation  | None                                                                                      |

## 15.5 Relationships

| Field         | Property    | Value                                                                                            |
|---------------|-------------|--------------------------------------------------------------------------------------------------|
| CONTRIBUTOR   | Type        | TEXT                                                                                             |
|               | Description | ID in <code>Person</code> table of the contributor                                               |
|               | Standards   | dc:contributor                                                                                   |
|               | Validation  | Must be an ID of a <code>Person</code>                                                           |
|               | Null        | Not null                                                                                         |
| DOCUMENTATION | Automation  | None                                                                                             |
|               | Type        | TEXT                                                                                             |
|               | Description | ID in <code>Documentation</code> table if the contribution relates to <code>Documentation</code> |
|               | Standards   | dc:relation                                                                                      |
|               | Validation  | Must be an ID of <code>Documentation</code>                                                      |
| APPLICATION   | Null        | Null if <code>APPLICATION</code> is not null                                                     |
|               | Automation  | None                                                                                             |
|               | Type        | TEXT                                                                                             |
|               | Description | ID in <code>Application</code> table if the contribution relates to an <code>Application</code>  |
|               | Standards   | dc:relation                                                                                      |
|               | Validation  | Must be an ID of an <code>Application</code>                                                     |
|               | Null        | Null if <code>DOCUMENTATION</code> is not null                                                   |
|               | Automation  | None                                                                                             |

## 16 Dependency

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 16.1 Description

The `Dependency` table records that one `Application` depends on another, optionally under certain conditions.

### 16.2 Standards

None

### 16.3 Automation

A `Dependency` may be inferred automatically in some cases, but are generally provided by the user.

### 16.4 Attributes

| Field       | Property    | Value                                                        |
|-------------|-------------|--------------------------------------------------------------|
| OPTIONALITY | Type        | TEXT                                                         |
|             | Description | Whether the dependency is required, optional, or conditional |
|             | Standards   | None                                                         |
|             | Validation  | One of 'required', 'optional', or a condition string         |
|             | Automation  | None                                                         |

## 16.5 Relationships

| Field     | Property    | Value                                                                                        |
|-----------|-------------|----------------------------------------------------------------------------------------------|
| DEPENDANT | Type        | TEXT                                                                                         |
|           | Description | ID in <code>Application</code> table of the <code>Application</code> that depends on another |
|           | Standards   | None                                                                                         |
|           | Validation  | Must be an ID of an <code>Application</code>                                                 |

| Field      | Property    | Value                                                          |
|------------|-------------|----------------------------------------------------------------|
| DEPENDENCY | Null        | Not null                                                       |
|            | Automation  | None                                                           |
|            | Type        | TEXT                                                           |
|            | Description | ID in Application table of the Application being depended upon |
|            | Standards   | None                                                           |
|            | Validation  | Must be an ID of an Application                                |
|            | Null        | Not null                                                       |
|            | Automation  | None                                                           |

## 17 Documentation

### 17.1 Description

The Documentation table records documents that describe Applications or Studies.

### 17.2 Standards

dc:title; dc:created

### 17.3 Automation

None

### 17.4 Attributes

| Field            | Property    | Value                                    |
|------------------|-------------|------------------------------------------|
| ID_DOCUMENTATION | Type        | TEXT                                     |
|                  | Description | Unique identifier for the Documentation  |
|                  | Standards   | None                                     |
|                  | Validation  | Must be unique                           |
|                  | Automation  | Automated by the instantiating framework |
| TITLE            | Type        | TEXT                                     |
|                  | Description | Title of the documentation               |
|                  | Standards   | dc:title                                 |
|                  | Validation  | String                                   |
|                  | Automation  | None                                     |
| DATE             | Type        | TEXT                                     |
|                  | Description | Date the documentation was created       |
|                  | Standards   | dc:created; ISO8601                      |
|                  | Validation  | Datetime string                          |
|                  | Automation  | None                                     |

### 17.5 Relationships

| Field     | Property    | Value                                                                   |
|-----------|-------------|-------------------------------------------------------------------------|
| DOCUMENTS | Type        | TEXT                                                                    |
|           | Description | ID in Application table of the Application this Documentation describes |
|           | Standards   | dc:relation                                                             |
|           | Validation  | Must be an ID of an Application                                         |
|           | Null        | Null if REFERENCES is not null                                          |
| DESCRIBES | Automation  | None                                                                    |
|           | Type        | TEXT                                                                    |
|           | Description | ID in Study table of the Study this Documentation references            |
|           | Standards   | dc:relation                                                             |
|           | Validation  | Must be an ID of a Study                                                |

| Field | Property   | Value                         |
|-------|------------|-------------------------------|
|       | Null       | Null if DOCUMENTS is not null |
|       | Automation | None                          |

## 18 Employs

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 18.1 Description

The `Employs` table records that a `StatisticalMethod` or `VisualisationMethod` employs a `StatisticalVariable`.

### 18.2 Standards

None

### 18.3 Automation

This relationship may be inferred automatically from the definitions of methods and variables.

### 18.4 Relationships

| Field                | Property    | Value                                                                                            |
|----------------------|-------------|--------------------------------------------------------------------------------------------------|
| STATISTICAL_METHOD   | Type        | TEXT                                                                                             |
|                      | Description | ID in <code>StatisticalMethod</code> table if the relationship involves a statistical method     |
|                      | Standards   | None                                                                                             |
|                      | Validation  | Must be an ID of a <code>StatisticalMethod</code>                                                |
|                      | Null        | Null if <code>VISUALISATION_METHOD</code> is not null                                            |
|                      | Automation  | None                                                                                             |
| VISUALISATION_METHOD | Type        | TEXT                                                                                             |
|                      | Description | ID in <code>VisualisationMethod</code> table if the relationship involves a visualisation method |
|                      | Standards   | None                                                                                             |
|                      | Validation  | Must be an ID of a <code>VisualisationMethod</code>                                              |
|                      | Null        | Null if <code>STATISTICAL_METHOD</code> is not null                                              |
|                      | Automation  | None                                                                                             |
| STATISTICAL_VARIABLE | Type        | TEXT                                                                                             |
|                      | Description | ID in <code>StatisticalVariable</code> table of the variable being employed                      |
|                      | Standards   | None                                                                                             |
|                      | Validation  | Must be an ID of a <code>StatisticalVariable</code>                                              |
|                      | Null        | Not null                                                                                         |
|                      | Automation  | None                                                                                             |

## 19 Entailment

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 19.1 Description

The `Entailment` table records that a `StatisticalMethod` or `VisualisationMethod` entails an `Assumption`.

## 19.2 Standards

None

## 19.3 Automation

This relationship may be inferred automatically from the definitions of methods and assumptions.

## 19.4 Relationships

| Field                | Property    | Value                                                                                            |
|----------------------|-------------|--------------------------------------------------------------------------------------------------|
| STATISTICAL_METHOD   | Type        | TEXT                                                                                             |
|                      | Description | ID in <code>StatisticalMethod</code> table if the relationship involves a statistical method     |
|                      | Standards   | None                                                                                             |
|                      | Validation  | Must be an ID of a <code>StatisticalMethod</code>                                                |
|                      | Null        | Null if <code>VISUALISATION_METHOD</code> is not null                                            |
| VISUALISATION_METHOD | Automation  | None                                                                                             |
|                      | Type        | TEXT                                                                                             |
|                      | Description | ID in <code>VisualisationMethod</code> table if the relationship involves a visualisation method |
|                      | Standards   | None                                                                                             |
|                      | Validation  | Must be an ID of a <code>VisualisationMethod</code>                                              |
| ASSUMPTION           | Null        | Null if <code>STATISTICAL_METHOD</code> is not null                                              |
|                      | Automation  | None                                                                                             |
|                      | Type        | TEXT                                                                                             |
|                      | Description | ID in <code>Assumption</code> table of the <code>Assumption</code> that is entailed              |
|                      | Standards   | None                                                                                             |
|                      | Validation  | Must be an ID of an <code>Assumption</code>                                                      |
|                      | Null        | Not null                                                                                         |
|                      | Automation  | None                                                                                             |

## 20 Implements

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 20.1 Description

The `Implements` table records that an `Application` implements a `StatisticalMethod` or `VisualisationMethod`.

### 20.2 Standards

None

### 20.3 Automation

None

### 20.4 Attributes

| Field    | Property    | Value                                                                               |
|----------|-------------|-------------------------------------------------------------------------------------|
| FUNCTION | Type        | TEXT                                                                                |
|          | Description | Name of the function within the <code>Application</code> that implements the method |
|          | Standards   | None                                                                                |

| Field   | Property    | Value                                                  |
|---------|-------------|--------------------------------------------------------|
| LIBRARY | Validation  | String                                                 |
|         | Automation  | None                                                   |
|         | Type        | TEXT                                                   |
|         | Description | Name of the library within which the function is found |
|         | Standards   | None                                                   |
|         | Validation  | String                                                 |
|         | Automation  | None                                                   |

## 20.5 Relationships

| Field                | Property    | Value                                                                               |
|----------------------|-------------|-------------------------------------------------------------------------------------|
| STATISTICAL_METHOD   | Type        | TEXT                                                                                |
|                      | Description | ID in StatisticalMethod table if the implementation is for a statistical method     |
|                      | Standards   | None                                                                                |
|                      | Validation  | Must be an ID of a StatisticalMethod                                                |
|                      | Null        | Null if VISUALISATION_METHOD is not null                                            |
|                      | Automation  | None                                                                                |
| VISUALISATION_METHOD | Type        | TEXT                                                                                |
|                      | Description | ID in VisualisationMethod table if the implementation is for a visualisation method |
|                      | Standards   | None                                                                                |
|                      | Validation  | Must be an ID of a VisualisationMethod                                              |
|                      | Null        | Null if STATISTICAL_METHOD is not null                                              |
|                      | Automation  | None                                                                                |
| APPLICATION          | Type        | TEXT                                                                                |
|                      | Description | ID in Application table of the Application implementing the method                  |
|                      | Standards   | None                                                                                |
|                      | Validation  | Must be an ID of an Application                                                     |
|                      | Null        | Not null                                                                            |
|                      | Automation  | None                                                                                |

## 21 Input

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 21.1 Description

The Input table records the Boxes that are used as input to a Process, including how they are used.

### 21.2 Standards

PROV:used

### 21.3 Automation

Populated automatically when a Process is invoked and its inputs are identified.

### 21.4 Attributes

| Field | Property    | Value                                            |
|-------|-------------|--------------------------------------------------|
| USAGE | Type        | TEXT                                             |
|       | Description | Indicates how the input is used by the Process   |
|       | Standards   | None                                             |
|       | Validation  | One of 'dependency' or 'data'                    |
|       | Automation  | Set automatically based on the role of the input |

## 21.5 Relationships

| Field   | Property    | Value                                               |
|---------|-------------|-----------------------------------------------------|
| PROCESS | Type        | TEXT                                                |
|         | Description | ID in Process table of the Process using the input  |
|         | Standards   | PROV:used                                           |
|         | Validation  | Must be an ID of a Process                          |
|         | Null        | Not null                                            |
|         | Automation  | Populated automatically when the Process is created |
| BOX     | Type        | TEXT                                                |
|         | Description | ID in Box table of the Box being used as input      |
|         | Standards   | PROV:Entity                                         |
|         | Validation  | Must be an ID of a Box                              |
|         | Null        | Not null                                            |
|         | Automation  | Populated automatically when inputs are resolved    |

## 22 Involvement

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 22.1 Description

The *Involvement* table records the involvement of a *Person* in a *Study*, including their role.

### 22.2 Standards

PROV:wasAssociatedWith

### 22.3 Automation

None

### 22.4 Attributes

| Field | Property    | Value                           |
|-------|-------------|---------------------------------|
| ROLE  | Type        | TEXT                            |
|       | Description | Role of the Person in the Study |
|       | Standards   | None                            |
|       | Validation  | String                          |
|       | Automation  | None                            |

### 22.5 Relationships

| Field  | Property    | Value                                                                                           |
|--------|-------------|-------------------------------------------------------------------------------------------------|
| PERSON | Type        | TEXT                                                                                            |
|        | Description | ID in <code>Person</code> table of the <code>Person</code> involved                             |
|        | Standards   | PROV:Agent                                                                                      |
|        | Validation  | Must be an ID of a <code>Person</code>                                                          |
|        | Null        | Not null                                                                                        |
| STUDY  | Automation  | None                                                                                            |
|        | Type        | TEXT                                                                                            |
|        | Description | ID in <code>Study</code> table of the <code>Study</code> the <code>Person</code> is involved in |
|        | Standards   | PROV:Activity                                                                                   |
|        | Validation  | Must be an ID of a <code>Study</code>                                                           |
|        | Null        | Not null                                                                                        |
|        | Automation  | None                                                                                            |

## 23 Meets

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 23.1 Description

The `Meets` table records that a `Computer` meets a specified `Specification`.

### 23.2 Standards

None

### 23.3 Automation

None

### 23.4 Relationships

| Field                     | Property    | Value                                                                                                |
|---------------------------|-------------|------------------------------------------------------------------------------------------------------|
| COMPUTER_SPECIFICATION    | Type        | TEXT                                                                                                 |
|                           | Description | ID in <code>Specification</code> table describing the specification met by the <code>Computer</code> |
|                           | Standards   | None                                                                                                 |
|                           | Validation  | Must be an ID of a <code>Specification</code>                                                        |
|                           | Null        | Not null                                                                                             |
|                           | Automation  | None                                                                                                 |
| REQUIREMENT_SPECIFICATION | Type        | TEXT                                                                                                 |
|                           | Description | ID in <code>Specification</code> table defining the required specification being met                 |
|                           | Standards   | None                                                                                                 |
|                           | Validation  | Must be an ID of a <code>Specification</code>                                                        |
|                           | Null        | Not null                                                                                             |
|                           | Automation  | None                                                                                                 |

## 24 Model

### 24.1 Description

The `Model` table identifies `Applications` that represent models, optionally linking to external model registries.

## 24.2 Standards

None

## 24.3 Automation

None

## 24.4 Attributes

| Field    | Property    | Value                                                    |
|----------|-------------|----------------------------------------------------------|
| ID_MODEL | Type        | TEXT                                                     |
|          | Description | Unique identifier for the Model                          |
|          | Standards   | None                                                     |
|          | Validation  | Must be unique                                           |
|          | Automation  | Automated by the instantiating framework                 |
| COMSES   | Type        | TEXT                                                     |
|          | Description | Reference to an entry in the CoMSES-Net model repository |
|          | Standards   | None                                                     |
|          | Validation  | String or identifier                                     |
|          | Null        | Optional                                                 |
|          | Automation  | None                                                     |

## 24.5 Relationships

| Field       | Property    | Value                                                      |
|-------------|-------------|------------------------------------------------------------|
| APPLICATION | Type        | TEXT                                                       |
|             | Description | ID in Application table of the Application that is a Model |
|             | Standards   | None                                                       |
|             | Validation  | Must be an ID of an Application                            |
|             | Null        | Not null                                                   |
|             | Automation  | None                                                       |

# 25 Parameter

## 25.1 Description

The Parameter table records parameters used by StatisticalMethod or VisualisationMethod.

## 25.2 Standards

None

## 25.3 Automation

None

## 25.4 Attributes

| Field        | Property    | Value                               |
|--------------|-------------|-------------------------------------|
| ID_PARAMETER | Type        | TEXT                                |
|              | Description | Unique identifier for the Parameter |
|              | Standards   | None                                |
|              | Validation  | Must be unique                      |

| Field     | Property    | Value                                    |
|-----------|-------------|------------------------------------------|
| DATA_TYPE | Automation  | Automated by the instantiating framework |
|           | Type        | TEXT                                     |
|           | Description | Data type of the parameter               |
|           | Standards   | XSD                                      |
|           | Validation  | Valid XSD data type or URI               |
|           | Automation  | None                                     |

## 25.5 Relationships

| Field                | Property    | Value                                                                                           |
|----------------------|-------------|-------------------------------------------------------------------------------------------------|
| STATISTICAL_METHOD   | Type        | TEXT                                                                                            |
|                      | Description | ID in <code>StatisticalMethod</code> table if the parameter applies to a statistical method     |
|                      | Standards   | None                                                                                            |
|                      | Validation  | Must be an ID of a <code>StatisticalMethod</code>                                               |
|                      | Null        | Null if <code>VISUALISATION_METHOD</code> is not null                                           |
|                      | Automation  | None                                                                                            |
| VISUALISATION_METHOD | Type        | TEXT                                                                                            |
|                      | Description | ID in <code>VisualisationMethod</code> table if the parameter applies to a visualisation method |
|                      | Standards   | None                                                                                            |
|                      | Validation  | Must be an ID of a <code>VisualisationMethod</code>                                             |
|                      | Null        | Null if <code>STATISTICAL_METHOD</code> is not null                                             |
|                      | Automation  | None                                                                                            |

## 26 Person

### 26.1 Description

The `Person` table records individuals associated with the system, such as users, contributors, or data owners.

### 26.2 Standards

PROV:Agent; FOAF

### 26.3 Automation

None

### 26.4 Attributes

| Field     | Property    | Value                                         |
|-----------|-------------|-----------------------------------------------|
| ID_PERSON | Type        | TEXT                                          |
|           | Description | Unique identifier for the <code>Person</code> |
|           | Standards   | FOAF                                          |
|           | Validation  | Must be unique                                |
|           | Automation  | Automated by the instantiating framework      |
| EMAIL     | Type        | TEXT                                          |
|           | Description | Email address of the <code>Person</code>      |
|           | Standards   | FOAF                                          |
|           | Validation  | Valid email address string                    |
|           | Automation  | None                                          |

## 27 PersonalData

### 27.1 Description

The `PersonalData` table records additional pieces of information about a `Person` as label-value pairs.

### 27.2 Standards

None

### 27.3 Automation

None

### 27.4 Attributes

| Field            | Property    | Value                                                     |
|------------------|-------------|-----------------------------------------------------------|
| ID_PERSONAL_DATA | Type        | TEXT                                                      |
|                  | Description | Unique identifier for the <code>PersonalData</code> entry |
|                  | Standards   | None                                                      |
|                  | Validation  | Must be unique                                            |
|                  | Automation  | Automated by the instantiating framework                  |
| CATEGORY         | Type        | TEXT                                                      |
|                  | Description | Category describing the type of personal data             |
|                  | Standards   | None                                                      |
|                  | Validation  | String                                                    |
|                  | Automation  | None                                                      |
| VALUE            | Type        | TEXT                                                      |
|                  | Description | Value of the personal data                                |
|                  | Standards   | None                                                      |
|                  | Validation  | String                                                    |
|                  | Automation  | None                                                      |

### 27.5 Relationships

| Field | Property    | Value                                                                         |
|-------|-------------|-------------------------------------------------------------------------------|
| ABOUT | Type        | TEXT                                                                          |
|       | Description | ID in <code>Person</code> table of the <code>Person</code> this data is about |
|       | Standards   | None                                                                          |
|       | Validation  | Must be an ID of a <code>Person</code>                                        |
|       | Null        | Not null                                                                      |
|       | Automation  | None                                                                          |

## 28 Pipeline

### 28.1 Description

The `Pipeline` table records sequences of `Applications`, allowing workflows to be defined where one `Application` calls another.

### 28.2 Standards

None

## 28.3 Automation

None

## 28.4 Attributes

| Field       | Property    | Value                                    |
|-------------|-------------|------------------------------------------|
| ID_PIPELINE | Type        | TEXT                                     |
|             | Description | Unique identifier for the Pipeline       |
|             | Standards   | None                                     |
|             | Validation  | Must be unique                           |
|             | Automation  | Automated by the instantiating framework |

## 28.5 Relationships

| Field    | Property    | Value                                                                   |
|----------|-------------|-------------------------------------------------------------------------|
| CALLS    | Type        | TEXT                                                                    |
|          | Description | ID in Application table of the Application called by this Pipeline step |
|          | Standards   | None                                                                    |
|          | Validation  | Must be an ID of an Application                                         |
|          | Null        | Not null                                                                |
| PREVIOUS | Automation  | None                                                                    |
|          | Type        | TEXT                                                                    |
|          | Nullable    | True                                                                    |
|          | Description | ID in Pipeline table of the previous Pipeline step                      |
|          | Standards   | None                                                                    |
|          | Validation  | Must be an ID of a Pipeline                                             |
|          | Null        | Null if this is the first step                                          |
|          | Automation  | None                                                                    |

# 29 Process

## 29.1 Description

The Process table records an execution of an Application, including when and how it was run.

## 29.2 Standards

PROV:Activity

## 29.3 Automation

Entries are created automatically when an Application is executed, capturing runtime details.

## 29.4 Attributes

| Field      | Property    | Value                                    |
|------------|-------------|------------------------------------------|
| ID_PROCESS | Type        | TEXT                                     |
|            | Description | Unique identifier for the Process        |
|            | Standards   | None                                     |
|            | Validation  | Must be unique                           |
|            | Automation  | Automated by the instantiating framework |
| START_TIME | Type        | TEXT                                     |
|            | Description | Time at which the Process started        |
|            | Standards   | ISO8601                                  |

| Field       | Property    | Value                                                     |
|-------------|-------------|-----------------------------------------------------------|
| END_TIME    | Validation  | Datetime string                                           |
|             | Automation  | Captured automatically at process start                   |
|             | Type        | TEXT                                                      |
|             | Description | Time at which the Process ended                           |
| ARGV        | Standards   | ISO8601                                                   |
|             | Validation  | Datetime string                                           |
|             | Automation  | Captured automatically at process completion              |
|             | Type        | TEXT                                                      |
| ENVIRONMENT | Description | Command-line string used to invoke the Application        |
|             | Standards   | POSIX                                                     |
|             | Validation  | String                                                    |
|             | Automation  | Constructed automatically from Argument and ArgumentValue |
| WORKING_DIR | Type        | TEXT                                                      |
|             | Description | Environment variables used during execution               |
|             | Standards   | POSIX                                                     |
|             | Validation  | List of strings                                           |
|             | Automation  | Captured from runtime environment                         |
|             | Type        | TEXT                                                      |
|             | Description | Working directory from which the Process was executed     |
|             | Standards   | POSIX                                                     |
|             | Validation  | Valid file path                                           |
|             | Automation  | Captured automatically from runtime                       |

## 29.5 Relationships

| Field      | Property    | Value                                                        |
|------------|-------------|--------------------------------------------------------------|
| EXECUTABLE | Type        | TEXT                                                         |
|            | Description | ID in Application table of the Application that was executed |
|            | Standards   | PROV:used                                                    |
|            | Validation  | Must be an ID of an Application                              |
| SOME_USER  | Null        | Not null                                                     |
|            | Automation  | Set when the Process is created                              |
|            | Type        | TEXT                                                         |
|            | Description | ID in User table of the user who initiated the Process       |
| HOST       | Standards   | PROV:wasAssociatedWith                                       |
|            | Validation  | Must be an ID of a User                                      |
|            | Null        | Not null                                                     |
|            | Automation  | Captured from system user context                            |
| PARENT     | Type        | TEXT                                                         |
|            | Description | ID in Computer table of the machine on which the Process ran |
|            | Standards   | PROV:wasAssociatedWith                                       |
|            | Validation  | Must be an ID of a Computer                                  |
|            | Null        | Not null                                                     |
|            | Automation  | Captured from system environment                             |
|            | Type        | TEXT                                                         |
|            | Description | ID in Process table of the parent Process, if any            |
|            | Standards   | PROV:wasInformedBy                                           |
|            | Validation  | Must be an ID of a Process                                   |
|            | Null        | Null if no parent process                                    |
|            | Automation  | Set if Process is spawned by another                         |

## 30 Product

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 30.1 Description

The Product table describes the types of output that an Application can produce, including how those outputs are located.

## 30.2 Standards

None

## 30.3 Automation

Some Products may be inferred automatically based on Application execution and generated Boxes.

## 30.4 Attributes

| Field       | Property    | Value                                                                   |
|-------------|-------------|-------------------------------------------------------------------------|
| OPTIONALITY | Type        | TEXT                                                                    |
|             | Description | Whether the Product is always produced or only under certain conditions |
|             | Standards   | None                                                                    |
|             | Validation  | One of 'always' or 'depends'                                            |
|             | Automation  | None                                                                    |
| LOCATOR     | Type        | TEXT                                                                    |
|             | Description | Description of how to locate the Product output                         |
|             | Standards   | None                                                                    |
|             | Validation  | Formatted rule string                                                   |
|             | Automation  | None                                                                    |

## 30.5 Relationships

| Field       | Property    | Value                                                               |
|-------------|-------------|---------------------------------------------------------------------|
| APPLICATION | Type        | TEXT                                                                |
|             | Description | ID in Application table of the Application producing this Product   |
|             | Standards   | None                                                                |
|             | Validation  | Must be an ID of an Application                                     |
|             | Null        | Not null                                                            |
|             | Automation  | None                                                                |
|             |             |                                                                     |
| BOX_TYPE    | Type        | TEXT                                                                |
|             | Description | ID in BoxType table describing the type of Box produced             |
|             | Standards   | None                                                                |
|             | Validation  | Must be an ID of a BoxType                                          |
|             | Null        | Not null                                                            |
|             | Automation  | None                                                                |
|             |             |                                                                     |
| IN_FILE     | Type        | TEXT                                                                |
|             | Description | ID in BoxType table if the Product is contained within another file |
|             | Standards   | None                                                                |
|             | Validation  | Must be an ID of a BoxType                                          |
|             | Null        | Null if LOCATOR is not 'in-file'                                    |
|             | Automation  | None                                                                |
|             |             |                                                                     |

# 31 Project

## 31.1 Description

The Project table records Projects, which are collections of Studies and may include funding and organisational information.

## 31.2 Standards

None

## 31.3 Automation

None

## 31.4 Attributes

| Field      | Property    | Value                                       |
|------------|-------------|---------------------------------------------|
| ID_PROJECT | Type        | TEXT                                        |
|            | Description | Unique identifier for the Project           |
|            | Standards   | None                                        |
|            | Validation  | Must be unique                              |
|            | Automation  | Automated by the instantiating framework    |
| TITLE      | Type        | TEXT                                        |
|            | Description | Title of the Project                        |
|            | Standards   | dc:title                                    |
|            | Validation  | String                                      |
|            | Automation  | None                                        |
| FUNDER     | Type        | TEXT                                        |
|            | Description | Organisation or body funding the Project    |
|            | Standards   | dc:publisher                                |
|            | Validation  | String                                      |
|            | Automation  | None                                        |
| GRANT_ID   | Type        | TEXT                                        |
|            | Description | Identifier of the grant funding the Project |
|            | Standards   | None                                        |
|            | Validation  | String                                      |
|            | Automation  | None                                        |

## 31.5 Relationships

| Field | Property    | Value                                                    |
|-------|-------------|----------------------------------------------------------|
| STUDY | Type        | TEXT                                                     |
|       | Description | A Study is a piece of work at some level of aggregation. |
|       | Standards   | None                                                     |
|       | Validation  | An ID in the Study table.                                |
|       | Null        | Not Null                                                 |
|       | Automation  | None                                                     |

## 32 Requirement

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 32.1 Description

The Requirement table records requirements that an Application has with respect to Specifications, including exact, minimum, or matching constraints.

### 32.2 Standards

None

### 32.3 Automation

None

### 32.4 Relationships

| Field       | Property    | Value                                                               |
|-------------|-------------|---------------------------------------------------------------------|
| APPLICATION | Type        | TEXT                                                                |
|             | Description | ID in Application table of the Application that has the requirement |
|             | Standards   | None                                                                |
|             | Validation  | Must be an ID of an Application                                     |
|             | Null        | Not null                                                            |
| MATCH       | Automation  | None                                                                |
|             | Type        | TEXT                                                                |
|             | Description | ID in Specification table specifying values that must match         |
|             | Standards   | None                                                                |
|             | Validation  | Must be an ID of a Specification                                    |
| MINIMUM     | Null        | Null if EXACT or MINIMUM is used                                    |
|             | Automation  | None                                                                |
|             | Type        | TEXT                                                                |
|             | Description | ID in Specification table specifying minimum acceptable values      |
|             | Standards   | None                                                                |
| EXACT       | Validation  | Must be an ID of a Specification                                    |
|             | Null        | Null if EXACT or MATCH is used                                      |
|             | Automation  | None                                                                |
|             | Type        | TEXT                                                                |
|             | Description | ID in Specification table specifying exact required values          |
|             | Standards   | None                                                                |
|             | Validation  | Must be an ID of a Specification                                    |
|             | Null        | Null if MATCH or MINIMUM is used                                    |
|             | Automation  | None                                                                |
|             | Type        | TEXT                                                                |

## 33 Specification

### 33.1 Description

The Specification table defines labelled specification values, typically used to describe properties of Computers or requirements of Applications.

### 33.2 Standards

None

### 33.3 Automation

None

### 33.4 Attributes

| Field            | Property    | Value                                         |
|------------------|-------------|-----------------------------------------------|
| ID_SPECIFICATION | Type        | TEXT                                          |
|                  | Description | Unique identifier for the Specification       |
|                  | Standards   | None                                          |
|                  | Validation  | Must be unique                                |
|                  | Automation  | Automated by the instantiating framework      |
| CATEGORY         | Type        | TEXT                                          |
|                  | Description | Category describing the type of specification |
|                  | Standards   | None                                          |
|                  | Validation  | String                                        |
|                  | Automation  | None                                          |
| VALUE            | Type        | TEXT                                          |
|                  | Description | Value of the specification                    |
|                  | Standards   | None                                          |
|                  | Validation  | String                                        |
|                  | Automation  | None                                          |

| Field | Property | Value |
|-------|----------|-------|
|-------|----------|-------|

### 33.5 Relationships

| Field            | Property    | Value                                                             |
|------------------|-------------|-------------------------------------------------------------------|
| SPECIFICATION_OF | Type        | TEXT                                                              |
|                  | Description | ID in Computer table of the Computer this Specification describes |
|                  | Standards   | None                                                              |
|                  | Validation  | Must be an ID of a Computer                                       |
|                  | Null        | Optional                                                          |
|                  | Automation  | None                                                              |

## 34 StatisticalInput

This is reification. In the graph database, this will appear as an edge between two entites. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 34.1 Description

The `StatisticalInput` table records the input `Variables` or `Boxes` used by a `StatisticalMethod` when performing a statistical computation.

### 34.2 Standards

PROV:used

### 34.3 Automation

Inputs may be inferred automatically when a `Statistics` computation is executed.

### 34.4 Relationships

| Field         | Property    | Value                                                                                                 |
|---------------|-------------|-------------------------------------------------------------------------------------------------------|
| STATISTICS    | Type        | TEXT                                                                                                  |
|               | Description | ID in <code>StatisticalMethod</code> table of the method using the input                              |
|               | Standards   | PROV:used                                                                                             |
|               | Validation  | Must be an ID of a <code>StatisticalMethod</code>                                                     |
|               | Null        | Not null                                                                                              |
|               | Automation  | None                                                                                                  |
| VISUALISATION | Type        | TEXT                                                                                                  |
|               | Description | ID in <code>Visualisations</code> table of the <code>Visualisation</code> used as input to the method |
|               | Standards   | PROV:Entity                                                                                           |
|               | Validation  | Must be an ID of a <code>Visualisation</code>                                                         |
|               | Null        | Null if <code>VARIABLE</code> is not null                                                             |
|               | Automation  | Resolved automatically where possible                                                                 |
| VALUE         | Type        | TEXT                                                                                                  |
|               | Description | ID in <code>Value</code> table of the <code>Variable</code> used as input to the method               |
|               | Standards   | None                                                                                                  |
|               | Validation  | Must be an ID of a <code>Value</code>                                                                 |
|               | Null        | Null if <code>Value</code> is not null                                                                |
|               | Automation  | Resolved automatically from <code>Content</code> definitions                                          |

## 35 StatisticalMethod

### 35.1 Description

The `StatisticalMethod` table records statistical methods that may be applied to data to generate `StatisticalVariables`.

### 35.2 Standards

None

### 35.3 Automation

None

### 35.4 Attributes

| Field                 | Property    | Value                                                    |
|-----------------------|-------------|----------------------------------------------------------|
| ID_STATISTICAL_METHOD | Type        | TEXT                                                     |
|                       | Description | Unique identifier for the <code>StatisticalMethod</code> |
|                       | Standards   | None                                                     |
|                       | Validation  | Must be unique                                           |
|                       | Automation  | Automated by the instantiating framework                 |

## 36 StatisticalVariable

### 36.1 Description

The `StatisticalVariable` table records variables that are generated by applying a `StatisticalMethod`.

### 36.2 Standards

None

### 36.3 Automation

`StatisticalVariables` may be created automatically as outputs of statistical computations.

### 36.4 Attributes

| Field                   | Property    | Value                                                      |
|-------------------------|-------------|------------------------------------------------------------|
| ID_STATISTICAL_VARIABLE | Type        | TEXT                                                       |
|                         | Description | Unique identifier for the <code>StatisticalVariable</code> |
|                         | Standards   | None                                                       |
|                         | Validation  | Must be unique                                             |
|                         | Automation  | Automated by the instantiating framework                   |
| DATA_TYPE               | Type        | TEXT                                                       |
|                         | Description | Data type of the <code>StatisticalVariable</code>          |
|                         | Standards   | XSD                                                        |
|                         | Validation  | Valid XSD data type or URI                                 |
|                         | Automation  | None                                                       |

## 36.5 Relationships

| Field                      | Property    | Value                                                                                                         |
|----------------------------|-------------|---------------------------------------------------------------------------------------------------------------|
| STATISTIC_GENERATED_BY     | Type        | TEXT                                                                                                          |
|                            | Description | ID in <code>StatisticalMethod</code> table of the method that generates this <code>StatisticalVariable</code> |
|                            | Standards   | PROV:wasGeneratedBy                                                                                           |
|                            | Validation  | Must be an ID of a <code>StatisticalMethod</code>                                                             |
|                            | Null        | Not null                                                                                                      |
| VISUALISATION_GENERATED_BY | Automation  | None                                                                                                          |
|                            | Type        | TEXT                                                                                                          |
|                            | Description | ID in <code>VisualisationMethod</code> table of the method that generates this <code>Visualisation</code>     |
|                            | Standards   | PROV:wasGeneratedBy                                                                                           |
|                            | Validation  | Must be an ID of a <code>VisualisationMethod</code> .                                                         |
|                            | Null        | Not null                                                                                                      |
|                            | Automation  | None                                                                                                          |

## 37 Statistics

### 37.1 Description

The `Statistics` table records statistical computations, including when they were performed and how input data was selected.

### 37.2 Standards

PROV:Activity

### 37.3 Automation

Entries may be created automatically when statistical computations are performed.

### 37.4 Attributes

| Field         | Property    | Value                                                         |
|---------------|-------------|---------------------------------------------------------------|
| ID_STATISTICS | Type        | TEXT                                                          |
|               | Description | Unique identifier for the <code>Statistics</code> computation |
|               | Standards   | None                                                          |
|               | Validation  | Must be unique                                                |
|               | Automation  | Automated by the instantiating framework                      |
| DATE          | Type        | DATE                                                          |
|               | Description | Date the statistical computation was performed                |
|               | Standards   | ISO8601                                                       |
|               | Validation  | Datetime string                                               |
|               | Automation  | Set automatically at execution time                           |
| QUERY         | Type        | TEXT                                                          |
|               | Description | Query used to select the data for the statistical computation |
|               | Standards   | None                                                          |
|               | Validation  | Formatted string                                              |
|               | Automation  | None                                                          |

## 37.5 Relationships

| Field | Property    | Value                                                                             |
|-------|-------------|-----------------------------------------------------------------------------------|
| USED  | Type        | TEXT                                                                              |
|       | Description | ID in <code>StatisticalMethod</code> table of the method used for the computation |
|       | Standards   | PROV:used                                                                         |
|       | Validation  | Must be an ID of a <code>StatisticalMethod</code>                                 |
|       | Null        | Not null                                                                          |
|       | Automation  | Set when the <code>Statistics</code> entry is created                             |

## 38 Study

### 38.1 Description

The `Study` table records units of scientific work, representing collections of `Processes`, data, and outputs.

### 38.2 Standards

PROV:Activity

### 38.3 Automation

Some fields may be populated automatically when `Processes` are grouped into `Studies`.

### 38.4 Attributes

| Field      | Property    | Value                                              |
|------------|-------------|----------------------------------------------------|
| ID_STUDY   | Type        | TEXT                                               |
|            | Description | Unique identifier for the <code>Study</code>       |
|            | Standards   | None                                               |
|            | Validation  | Must be unique                                     |
|            | Automation  | Automated by the instantiating framework           |
| TITLE      | Type        | TEXT                                               |
|            | Description | Title or name of the <code>Study</code>            |
|            | Standards   | dc:title                                           |
|            | Validation  | String                                             |
|            | Automation  | None                                               |
| START_TIME | Type        | DATE                                               |
|            | Description | Start time of the <code>Study</code>               |
|            | Standards   | ISO8601                                            |
|            | Validation  | Datetime string                                    |
|            | Automation  | May be inferred from earliest <code>Process</code> |
| END_TIME   | Type        | DATE                                               |
|            | Description | End time of the <code>Study</code>                 |
|            | Standards   | ISO8601                                            |
|            | Validation  | Datetime string                                    |
|            | Automation  | May be inferred from latest <code>Process</code>   |

### 38.5 Relationships

| Field   | Property    | Value                                                                                           |
|---------|-------------|-------------------------------------------------------------------------------------------------|
| PROJECT | Type        | TEXT                                                                                            |
|         | Description | ID in <code>Project</code> table of the <code>Project</code> this <code>Study</code> belongs to |
|         | Standards   | None                                                                                            |
|         | Validation  | Must be an ID of a <code>Project</code>                                                         |
|         | Null        | Optional                                                                                        |
|         | Automation  | None                                                                                            |
| PART    | Type        | TEXT                                                                                            |
|         | Description | ID in <code>Study</code> table if this <code>Study</code> is part of another <code>Study</code> |

| Field | Property   | Value                             |
|-------|------------|-----------------------------------|
|       | Standards  | PROV:wasPartOf                    |
|       | Validation | Must be an ID of a Study          |
|       | Null       | Null if not part of another Study |
|       | Automation | None                              |

## 39 Tag

### 39.1 Description

The Tag table records tags that can be used to classify and annotate other entities in the system.

### 39.2 Standards

None

### 39.3 Automation

None

### 39.4 Attributes

| Field  | Property    | Value                                    |
|--------|-------------|------------------------------------------|
| ID_TAG | Type        | TEXT                                     |
|        | Description | Unique identifier for the Tag            |
|        | Standards   | None                                     |
|        | Validation  | Must be unique                           |
|        | Automation  | Automated by the instantiating framework |

## 40 TagMap

This is reification. In the graph database, this will appear as an edge between two entities. In a relational database, this will be a relation which allows the enumeration of the many-to-many linkage.

### 40.1 Description

The TagMap table records the association of Tags with other entities such as Applications, Boxes, Documentation, Studies, and Methods.

### 40.2 Standards

None

### 40.3 Automation

Tags may be applied manually by users or inferred automatically based on metadata and usage.

### 40.4 Relationships

| Field                | Property    | Value                                                                        |
|----------------------|-------------|------------------------------------------------------------------------------|
| TAG                  | Type        | TEXT                                                                         |
|                      | Description | ID in Tag table of the Tag being applied                                     |
|                      | Standards   | None                                                                         |
|                      | Validation  | Must be an ID of a Tag                                                       |
|                      | Null        | Not null                                                                     |
| APPLICATION          | Automation  | None                                                                         |
|                      | Type        | TEXT                                                                         |
|                      | Description | ID in Application table if the Tag applies to an Application                 |
|                      | Standards   | None                                                                         |
|                      | Validation  | Must be an ID of an Application                                              |
| ASSUMPTION           | Null        | Null if not applicable                                                       |
|                      | Automation  | None                                                                         |
|                      | Type        | TEXT                                                                         |
|                      | Description | ID in Assumptoin table if the Tag applies to an Assumption.                  |
|                      | Standards   | None                                                                         |
| BOX                  | Validation  | Must be an ID of an Assumption                                               |
|                      | Null        | Null if not applicable                                                       |
|                      | Automation  | None                                                                         |
|                      | Type        | TEXT                                                                         |
|                      | Description | ID in Box table if the Tag applies to a Box                                  |
| BOX_TYPE             | Standards   | None                                                                         |
|                      | Validation  | Must be an ID of a Box                                                       |
|                      | Null        | Null if not applicable                                                       |
|                      | Automation  | None                                                                         |
|                      | Type        | TEXT                                                                         |
| DOCUMENTATION        | Description | ID in BoxType table if the Tag applies to a BoxType                          |
|                      | Standards   | None                                                                         |
|                      | Validation  | Must be an ID of a BoxType                                                   |
|                      | Null        | Null if not applicable                                                       |
|                      | Automation  | None                                                                         |
| OTHER_TAG            | Type        | TEXT                                                                         |
|                      | Description | ID in Documentation table if the Tag applies to Documentation                |
|                      | Standards   | None                                                                         |
|                      | Validation  | Must be an ID of a Documentation                                             |
|                      | Null        | Null if not applicable                                                       |
| PERSON               | Automation  | None                                                                         |
|                      | Type        | TEXT                                                                         |
|                      | Description | ID in Tag table if the Tag relates to another Tag                            |
|                      | Standards   | None                                                                         |
|                      | Validation  | Must be an ID of a Tag                                                       |
| STATISTICAL_METHOD   | Null        | Null if not applicable                                                       |
|                      | Automation  | None                                                                         |
|                      | Type        | TEXT                                                                         |
|                      | Description | ID in Person table is the Person in the Person table.                        |
|                      | Standards   | None                                                                         |
| STUDY                | Validation  | Must be an ID of a Tag                                                       |
|                      | Null        | Null if not applicable                                                       |
|                      | Automation  | None                                                                         |
|                      | Type        | TEXT                                                                         |
|                      | Description | ID in Study table if the Tag applies to a Study                              |
| VISUALISATION_METHOD | Standards   | None                                                                         |
|                      | Validation  | Must be an ID of a Study                                                     |
|                      | Null        | Null if not applicable                                                       |
|                      | Automation  | None                                                                         |
|                      | Type        | TEXT                                                                         |
|                      | Description | ID in VisualisationMethod table if the Tag applies to a visualisation method |
|                      | Standards   | None                                                                         |
|                      | Validation  | Must be an ID of a VisualisationMethod                                       |

| Field | Property   | Value                  |
|-------|------------|------------------------|
|       | Null       | Null if not applicable |
|       | Automation | None                   |

## 41 User

### 41.1 Description

The `User` table records system user accounts, linking operating system user information to a `Person`.

### 41.2 Standards

PROV:Agent

### 41.3 Automation

Most values are obtained automatically from the operating system.

### 41.4 Attributes

| Field    | Property    | Value                                       |
|----------|-------------|---------------------------------------------|
| ID_USER  | Type        | TEXT                                        |
|          | Description | Unique identifier for the <code>User</code> |
|          | Standards   | None                                        |
|          | Validation  | Must be unique                              |
| HOME_DIR | Automation  | Automated by the instantiating framework    |
|          | Type        | TEXT                                        |
|          | Description | Home directory of the user                  |
|          | Standards   | POSIX                                       |
|          | Validation  | Valid file path                             |
|          | Automation  | Retrieved from the operating system         |

### 41.5 Relationships

| Field      | Property    | Value                                                                                                |
|------------|-------------|------------------------------------------------------------------------------------------------------|
| ACCOUNT_OF | Type        | TEXT                                                                                                 |
|            | Description | ID in <code>Person</code> table of the <code>Person</code> this <code>User</code> account belongs to |
|            | Standards   | PROV:actedOnBehalfOf                                                                                 |
|            | Validation  | Must be an ID of a <code>Person</code>                                                               |
|            | Null        | Not null                                                                                             |
|            | Automation  | Resolved from system/user configuration                                                              |

## 42 Uses

### 42.1 Description

The `Uses` table records that an `Application` uses a `BoxType` as an input or dependency.

### 42.2 Standards

PROV:used

## 42.3 Automation

May be inferred automatically based on Application execution and detected Inputs.

## 42.4 Attributes

| Field   | Property    | Value                                       |
|---------|-------------|---------------------------------------------|
| LOCATOR | Type        | TEXT                                        |
|         | Description | Description of how to locate the Uses input |
|         | Standards   | None                                        |
|         | Validation  | Formatted rule string                       |
|         | Automation  | None                                        |

## 42.5 Relationships

| Field       | Property    | Value                                                                  |
|-------------|-------------|------------------------------------------------------------------------|
| APPLICATION | Type        | TEXT                                                                   |
|             | Description | ID in Application table of the Application that uses the input         |
|             | Standards   | PROV:Activity                                                          |
|             | Validation  | Must be an ID of an Application                                        |
|             | Null        | Not null                                                               |
|             | Automation  | None                                                                   |
| BOX_TYPE    | Type        | TEXT                                                                   |
|             | Description | ID in BoxType table of the type of Box that is used                    |
|             | Standards   | PROV:Entity                                                            |
|             | Validation  | Must be an ID of a BoxType                                             |
|             | Null        | Not null                                                               |
|             | Automation  | None                                                                   |
| IN_FILE     | Type        | TEXT                                                                   |
|             | Description | ID in BoxType table if the Uses input is contained within another file |
|             | Standards   | None                                                                   |
|             | Validation  | Must be an ID of a BoxType                                             |
|             | Null        | Null if LOCATOR is not 'in-file'                                       |
|             | Automation  | None                                                                   |

# 43 Value

## 43.1 Description

The Value table represents values of Variables or StatisticalVariables. It is a virtual table whose entries are retrieved from Boxes rather than stored directly.

## 43.2 Standards

PROV:Entity

## 43.3 Automation

Values are not stored explicitly; they are retrieved dynamically from Boxes based on Content specifications.

## 43.4 Attributes

| Field    | Property   | Value       |
|----------|------------|-------------|
| ID_VALUE | Type       | TEXT        |
|          | Constraint | PRIMARY KEY |

| Field  | Property    | Value                                                                                                 |
|--------|-------------|-------------------------------------------------------------------------------------------------------|
| UNITS  | Description | Identifier or representation of the value                                                             |
|        | Standards   | None                                                                                                  |
|        | Validation  | This should be a unique string                                                                        |
|        | Automation  | Retrieved from underlying data in Box                                                                 |
|        | Type        | TEXT                                                                                                  |
| FORMAT | Description | The units of the value.                                                                               |
|        | Standards   | None                                                                                                  |
|        | Validation  | None                                                                                                  |
|        | Automation  | None.                                                                                                 |
|        | Type        | TEXT                                                                                                  |
|        | Description | ID in <code>Variable</code> table of the <code>Variable</code> this <code>Value</code> corresponds to |
|        | Standards   | None                                                                                                  |
|        | Validation  | Values are not always numbers.                                                                        |
|        | Automation  | None.                                                                                                 |

## 43.5 Relationships

| Field                   | Property    | Value                                                                                                              |
|-------------------------|-------------|--------------------------------------------------------------------------------------------------------------------|
| VARIABLE                | Type        | TEXT                                                                                                               |
|                         | Description | ID in <code>Variable</code> table of the <code>Variable</code> this <code>Value</code> corresponds to              |
|                         | Standards   | None                                                                                                               |
|                         | Validation  | Must be an ID of a <code>Variable</code>                                                                           |
|                         | Null        | Null if <code>STATISTICAL_VARIABLE</code> is used                                                                  |
| STATISTICAL_VARIABLE    | Automation  | Resolved via <code>Content</code> definitions                                                                      |
|                         | Type        | TEXT                                                                                                               |
|                         | Description | ID in <code>StatisticalVariable</code> table if this <code>Value</code> is the result of a statistical computation |
|                         | Standards   | None                                                                                                               |
|                         | Validation  | Must be an ID of a <code>StatisticalVariable</code>                                                                |
| PARAMETER               | Null        | Null if <code>VARIABLE</code> is used                                                                              |
|                         | Automation  | Resolved via statistical processing                                                                                |
|                         | Type        | TEXT                                                                                                               |
|                         | Description | ID in <code>Parameter</code> table if this <code>Value</code> corresponds to a parameter                           |
|                         | Standards   | None                                                                                                               |
| STATISTICAL_PARAMETER   | Validation  | Must be an ID of a <code>Parameter</code>                                                                          |
|                         | Null        | Optional                                                                                                           |
|                         | Automation  | Resolved during method execution                                                                                   |
|                         | Type        | TEXT                                                                                                               |
|                         | Description | ID in <code>Statistics</code> table if this <code>Value</code> corresponds to a <code>Statistic</code>             |
| VISUALISATION_PARAMETER | Standards   | None                                                                                                               |
|                         | Validation  | Must be an ID of a <code>Statistic</code>                                                                          |
|                         | Null        | Optional                                                                                                           |
|                         | Automation  | Resolved during method execution                                                                                   |
|                         | Type        | TEXT                                                                                                               |
| RESULT_OF               | Description | ID in the <code>Visualisation</code> table if this <code>Value</code> corresponds to a <code>Visualisation</code>  |
|                         | Standards   | None                                                                                                               |
|                         | Validation  | Must be an ID of a <code>Visualisation</code>                                                                      |
|                         | Null        | Optional                                                                                                           |
|                         | Automation  | Resolved during method execution                                                                                   |
| TIME                    | Type        | TEXT                                                                                                               |
|                         | Description | ID in <code>Statistics</code> table if this <code>Value</code> is the result of a statistical computation          |
|                         | Standards   | <code>PROV:wasGeneratedBy</code>                                                                                   |
|                         | Validation  | Must be an ID of a <code>Statistics</code>                                                                         |
|                         | Null        | Optional                                                                                                           |
|                         | Automation  | Set when statistical outputs are generated                                                                         |
|                         | Type        | TEXT                                                                                                               |
|                         | Description | ID in <code>Context</code> table representing the time associated with the <code>Value</code>                      |
|                         | Standards   | None                                                                                                               |
|                         | Validation  | Must be an ID of a <code>Context</code>                                                                            |
|                         | Null        | Optional                                                                                                           |

| Field        | Property    | Value                                                                         |
|--------------|-------------|-------------------------------------------------------------------------------|
| SPACE        | Automation  | Derived from Content locators                                                 |
|              | Type        | TEXT                                                                          |
|              | Description | ID in Context table representing the spatial context of the Value             |
|              | Standards   | None                                                                          |
|              | Validation  | Must be an ID of a Context                                                    |
| AGENT        | Null        | Optional                                                                      |
|              | Automation  | Derived from Content locators                                                 |
|              | Type        | TEXT                                                                          |
|              | Description | ID in Context table representing the agent associated with the Value          |
|              | Standards   | None                                                                          |
| LINK         | Validation  | Must be an ID of a Context                                                    |
|              | Null        | Optional                                                                      |
|              | Automation  | Derived from Content locators                                                 |
|              | Type        | TEXT                                                                          |
|              | Description | ID in Context table representing link relationships associated with the Value |
| CONTAINED_IN | Standards   | None                                                                          |
|              | Validation  | Must be an ID of a Context                                                    |
|              | Null        | Optional                                                                      |
|              | Automation  | Derived from Content locators                                                 |
|              | Type        | TEXT                                                                          |
|              | Description | ID in Box table from which the Value is retrieved                             |
|              | Standards   | PROV:Entity                                                                   |
|              | Validation  | Must be an ID of a Box                                                        |
|              | Null        | Not null                                                                      |
|              | Automation  | Determined from data source                                                   |

## 44 Variable

### 44.1 Description

The Variable table records variables that describe data, including their name, type, and roles such as time, space, agent, or link.

### 44.2 Standards

None

### 44.3 Automation

Variables are generally defined by the user; some roles may be inferred during data processing.

### 44.4 Attributes

| Field       | Property    | Value                      |
|-------------|-------------|----------------------------|
| ID_VARIABLE | Type        | TEXT                       |
|             | Description | Identifier of the variable |
|             | Standards   | None                       |
|             | Validation  | Must be unique             |
|             | Automation  | None                       |
| NAME        | Type        | TEXT                       |
|             | Description | Name of the variable       |
|             | Standards   | None                       |
|             | Validation  | String                     |
|             | Automation  | None                       |
| DATA_TYPE   | Type        | TEXT                       |
|             | Description | Data type of the variable  |
|             | Standards   | XSD                        |
|             | Validation  | Valid XSD data type or URI |
|             | Automation  | None                       |

| Field    | Property    | Value                                                           |
|----------|-------------|-----------------------------------------------------------------|
| IS_AGENT | Type        | INTEGER                                                         |
|          | Description | Indicates whether this variable represents an agent identifier  |
|          | Standards   | None                                                            |
|          | Validation  | True or False                                                   |
|          | Automation  | None                                                            |
| IS_LINK  | Type        | INTEGER                                                         |
|          | Description | Indicates whether this variable represents a link identifier    |
|          | Standards   | None                                                            |
|          | Validation  | True or False                                                   |
|          | Automation  | None                                                            |
| IS_SPACE | Type        | INTEGER                                                         |
|          | Description | Indicates whether this variable represents spatial information  |
|          | Standards   | None                                                            |
|          | Validation  | True or False                                                   |
|          | Automation  | None                                                            |
| IS_TIME  | Type        | INTEGER                                                         |
|          | Description | Indicates whether this variable represents temporal information |
|          | Standards   | None                                                            |
|          | Validation  | True or False                                                   |
|          | Automation  | None                                                            |

## 45 Visualisation

### 45.1 Description

The `Visualisation` table records visualisations generated from data, including when they were created and how the data was selected.

### 45.2 Standards

PROV:Activity

### 45.3 Automation

Entries may be created automatically when a visualisation is generated.

### 45.4 Attributes

| Field            | Property    | Value                                                        |
|------------------|-------------|--------------------------------------------------------------|
| ID_VISUALISATION | Type        | TEXT                                                         |
|                  | Description | Unique identifier for the <code>Visualisation</code>         |
|                  | Standards   | None                                                         |
|                  | Validation  | Must be unique                                               |
|                  | Automation  | Automated by the instantiating framework                     |
| DATE             | Type        | DATE                                                         |
|                  | Description | Date the <code>Visualisation</code> was created              |
|                  | Standards   | ISO8601                                                      |
|                  | Validation  | Datetime string                                              |
|                  | Automation  | Set automatically at creation time                           |
| QUERY            | Type        | TEXT                                                         |
|                  | Description | Query used to select data for the <code>Visualisation</code> |
|                  | Standards   | None                                                         |
|                  | Validation  | Formatted string                                             |
|                  | Automation  | None                                                         |

## 45.5 Relationships

| Field                | Property    | Value                                                                            |
|----------------------|-------------|----------------------------------------------------------------------------------|
| VISUALISATION_METHOD | Type        | TEXT                                                                             |
|                      | Description | ID in VisualisationMethod table of the method used to generate the Visualisation |
|                      | Standards   | PROV:used                                                                        |
|                      | Validation  | Must be an ID of a VisualisationMethod                                           |
|                      | Null        | Not null                                                                         |
| CONTAINED_IN         | Automation  | Set when the Visualisation is created                                            |
|                      | Type        | TEXT                                                                             |
|                      | Description | ID in Box table of the Box containing the Visualisation                          |
|                      | Standards   | PROV:Entity                                                                      |
|                      | Validation  | Must be an ID of a Box                                                           |
|                      | Null        | Optional                                                                         |
|                      | Automation  | Set if output is stored in a Box                                                 |

## 46 VisualisationMethod

### 46.1 Description

The VisualisationMethod table records methods used to generate visualisations from data.

### 46.2 Standards

None

### 46.3 Automation

None

### 46.4 Attributes

| Field                   | Property    | Value                                         |
|-------------------------|-------------|-----------------------------------------------|
| ID_VISUALISATION_METHOD | Type        | TEXT                                          |
|                         | Description | Unique identifier for the VisualisationMethod |
|                         | Standards   | None                                          |
|                         | Validation  | Must be unique                                |
|                         | Automation  | Automated by the instantiating framework      |

## 47 VisualisationValue

### 47.1 Description

The VisualisationValue table links a Value to a Visualisation, indicating which Values are used in a given Visualisation.

### 47.2 Standards

PROV:used

### 47.3 Automation

Entries are created automatically when a `Visualisation` is generated.

### 47.4 Relationships

| Field         | Property    | Value                                                                                                   |
|---------------|-------------|---------------------------------------------------------------------------------------------------------|
| VALUE         | Type        | TEXT                                                                                                    |
|               | Description | ID in <code>Value</code> table of the <code>Value</code> used in the <code>Visualisation</code> .       |
|               | Standards   | PROV:Entity                                                                                             |
|               | Validation  | Must be an ID of a <code>Value</code> .                                                                 |
|               | Null        | Not null.                                                                                               |
| VISUALISATION | Automation  | Populated automatically when the <code>Visualisation</code> is created.                                 |
|               | Type        | TEXT                                                                                                    |
|               | Description | ID in <code>Visualisation</code> table of the <code>Visualisation</code> using the <code>Value</code> . |
|               | Standards   | PROV:Activity.                                                                                          |
|               | Validation  | Must be an ID of a <code>Visualisation</code> .                                                         |
|               | Null        | Not null.                                                                                               |
|               | Automation  | Populated automatically when the <code>Visualisation</code> is created.                                 |

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